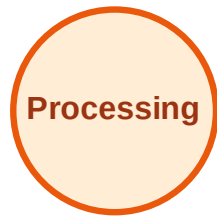


BFS Traversal

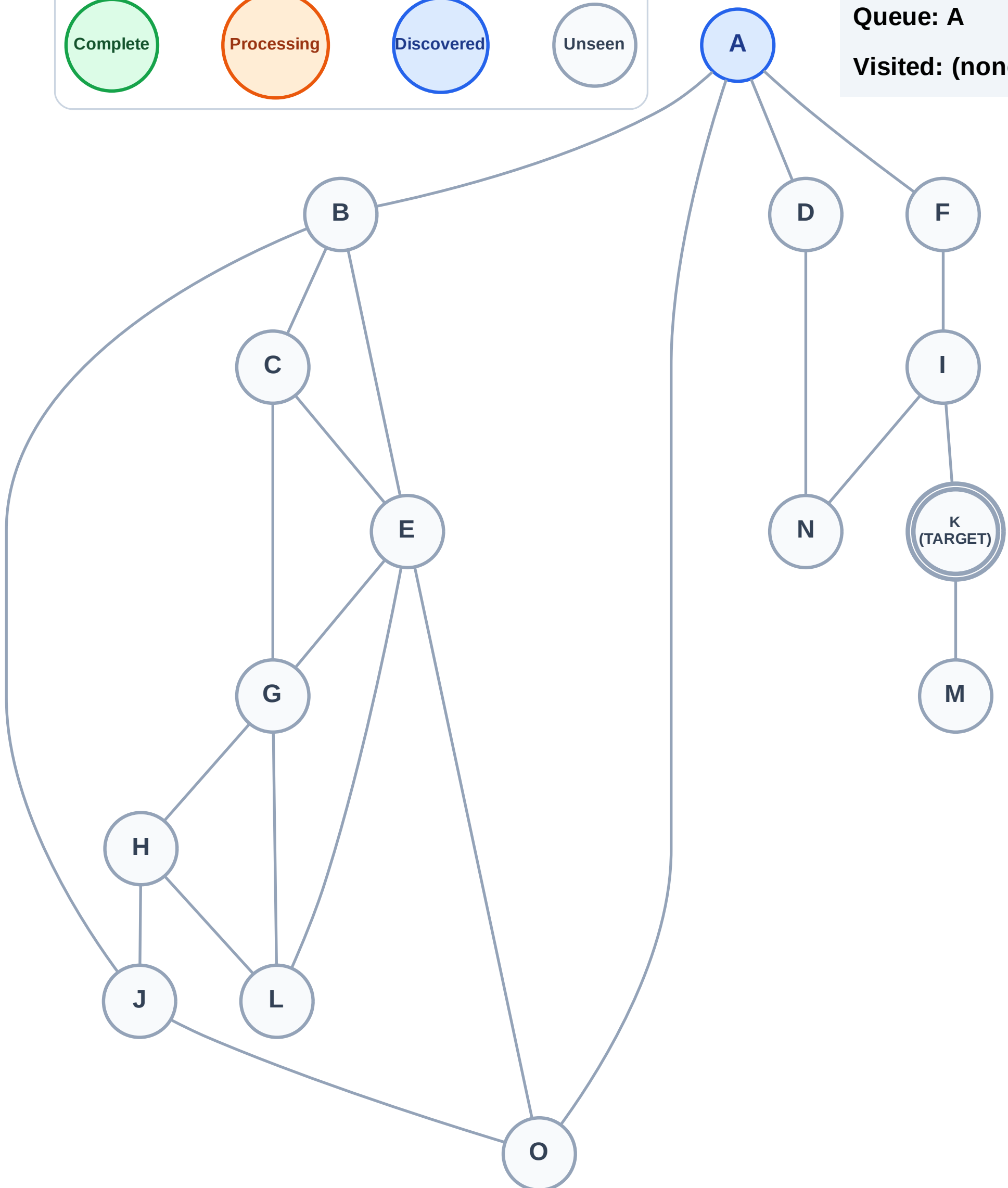
Step 1: Start at A (Seeking target node K)

Legend



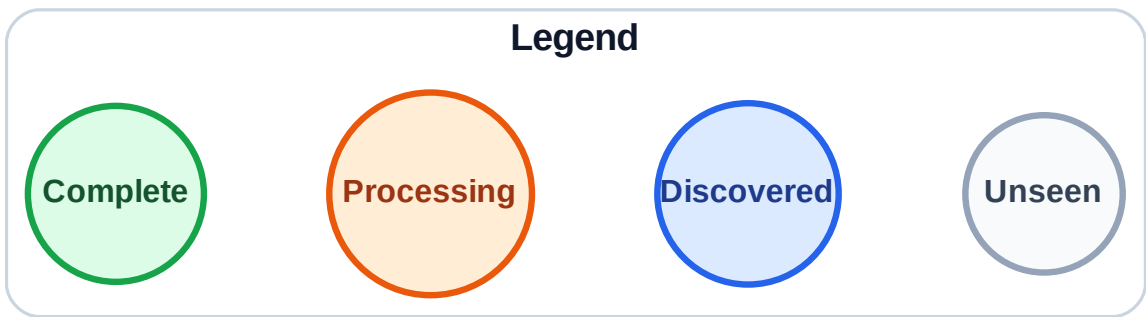
Queue: A

Visited: (none)

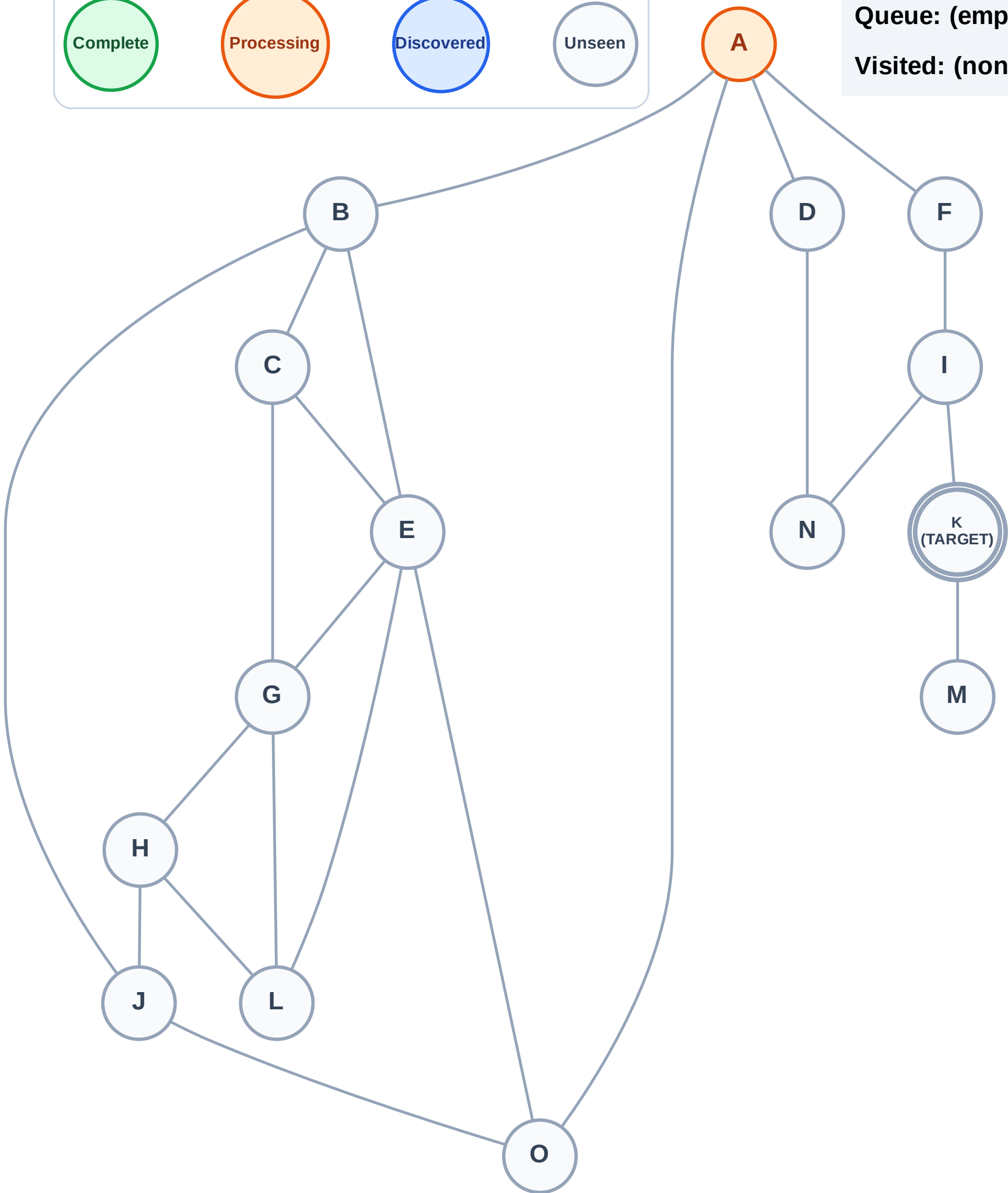


BFS Traversal

Step 2: Process node A



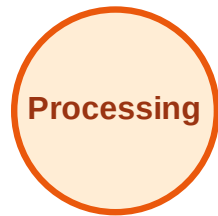
Queue: (empty)
Visited: (none)



BFS Traversal

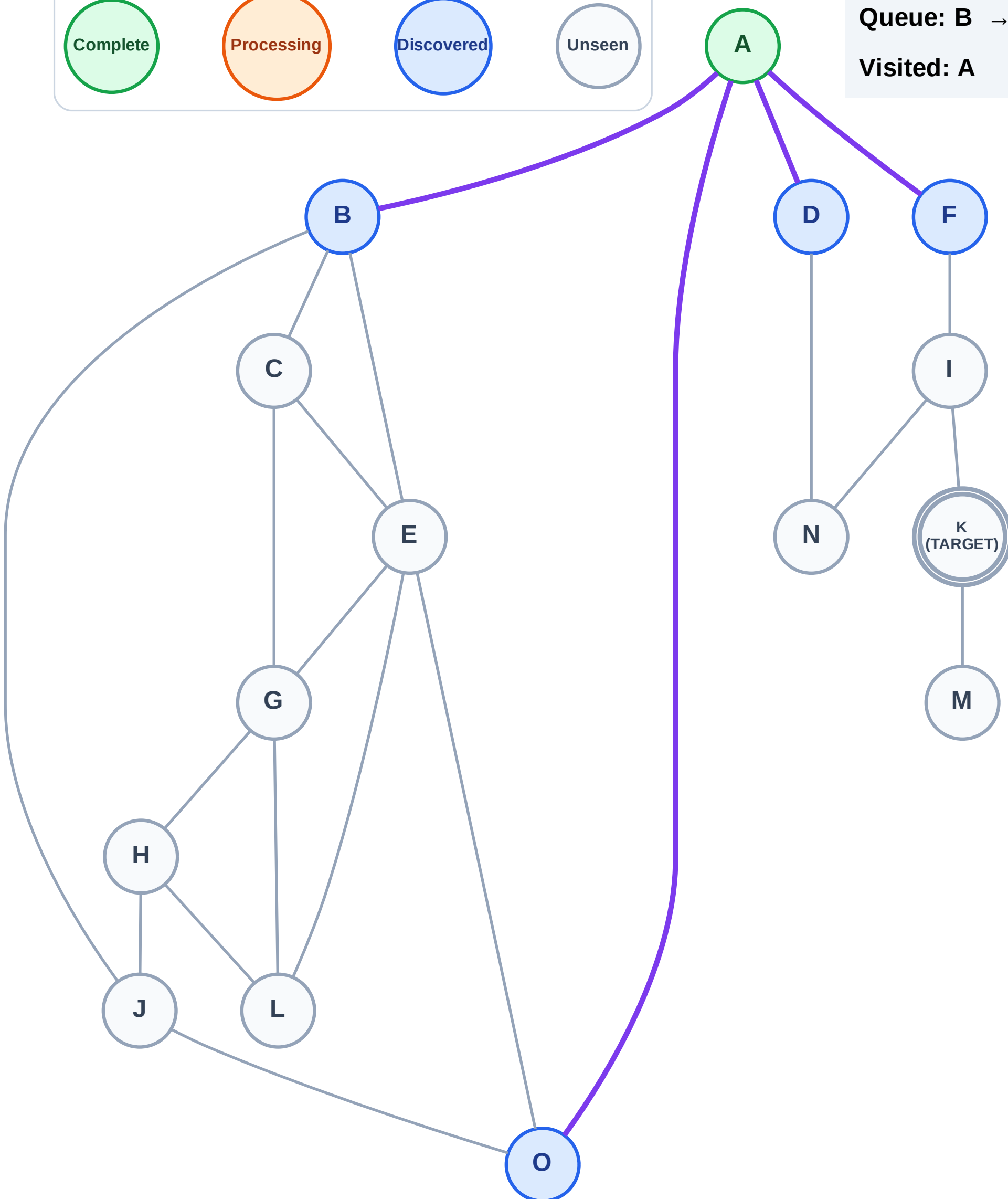
Step 3: Discover B, D, F, O from A

Legend



Queue: B → D → F → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

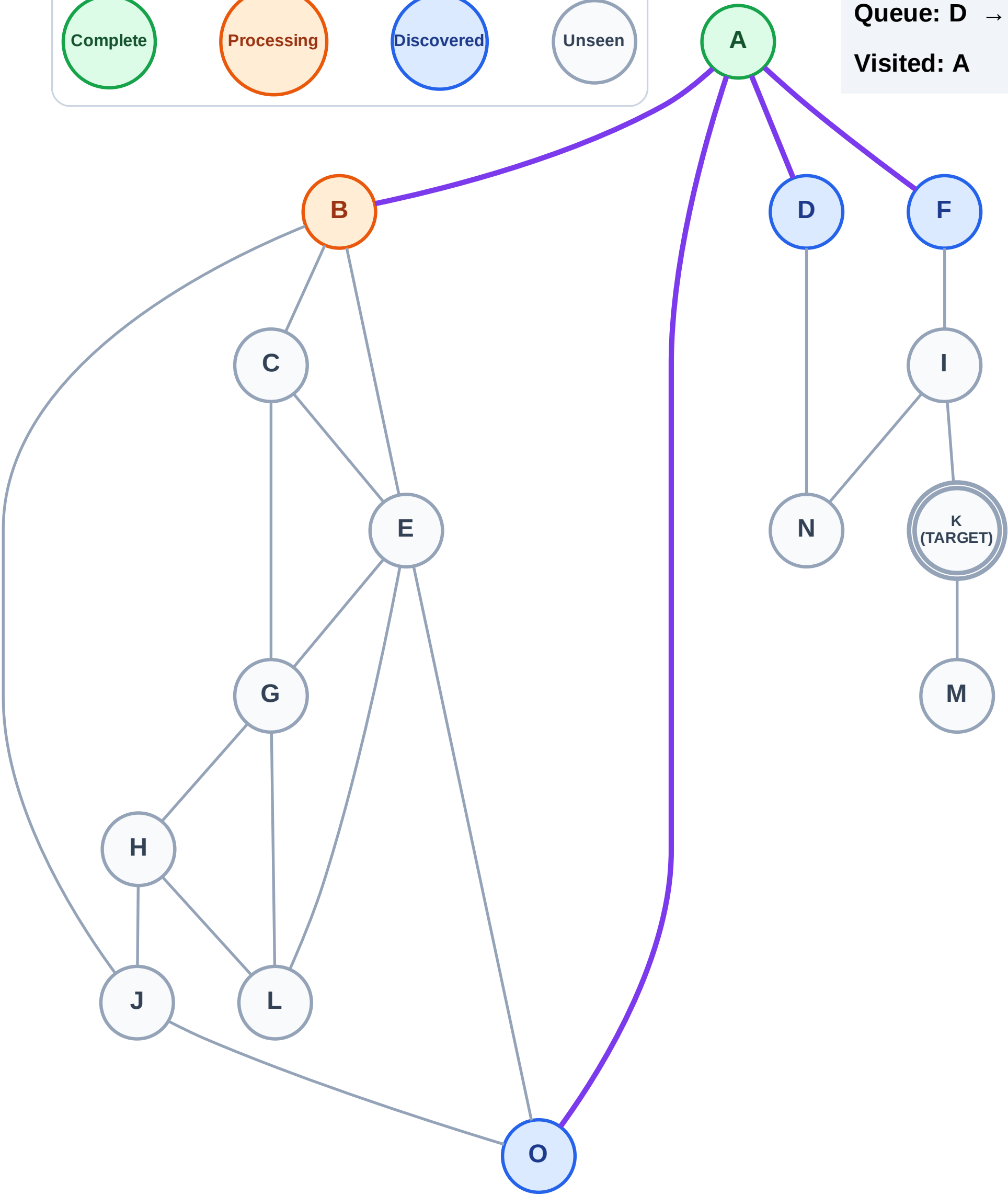
Processing

Discovered

Unseen

Queue: D → F → O

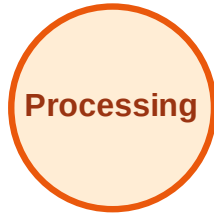
Visited: A



BFS Traversal

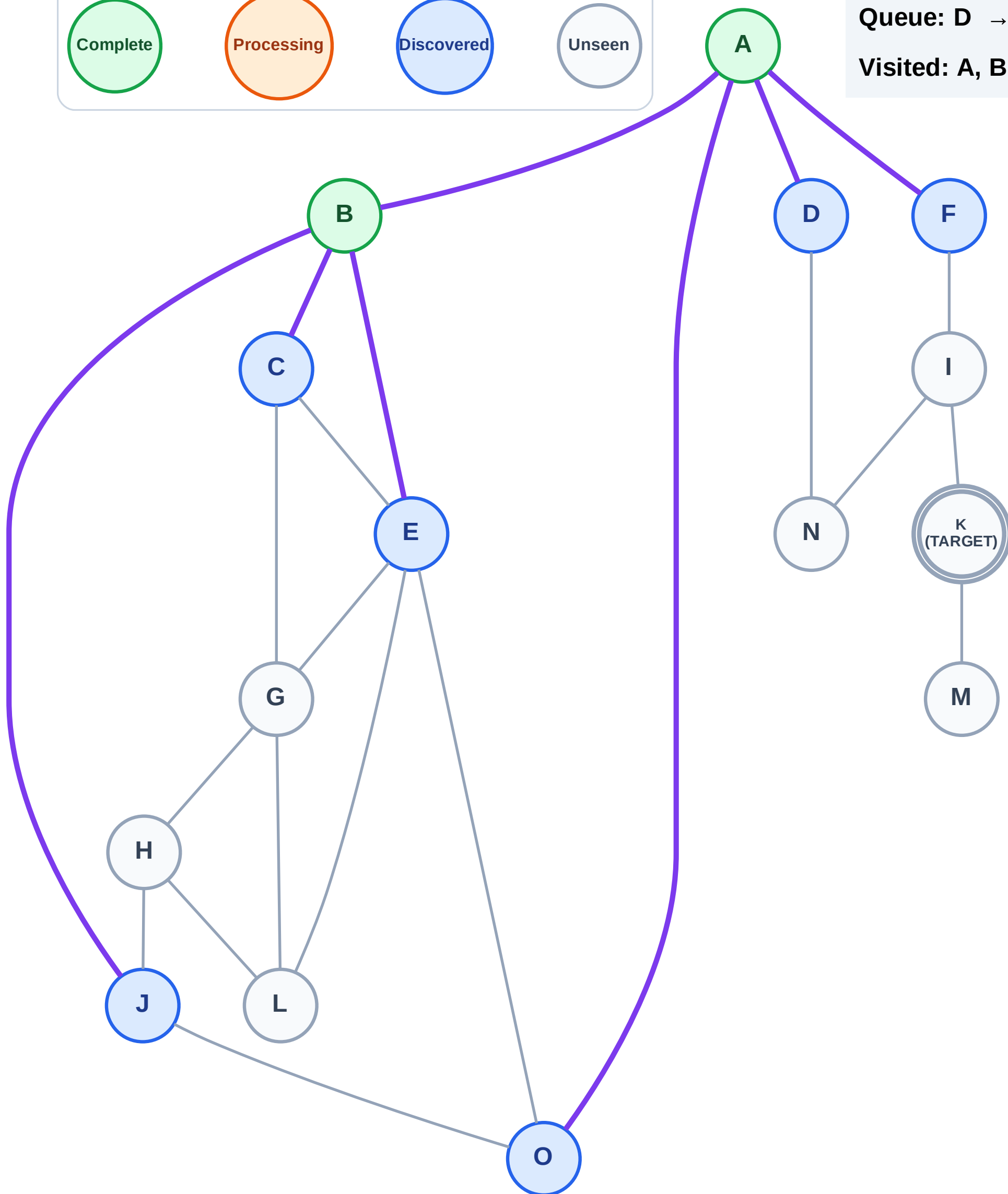
Step 5: Discover C, E, J from B

Legend



Queue: D → F → O → C → E → J

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

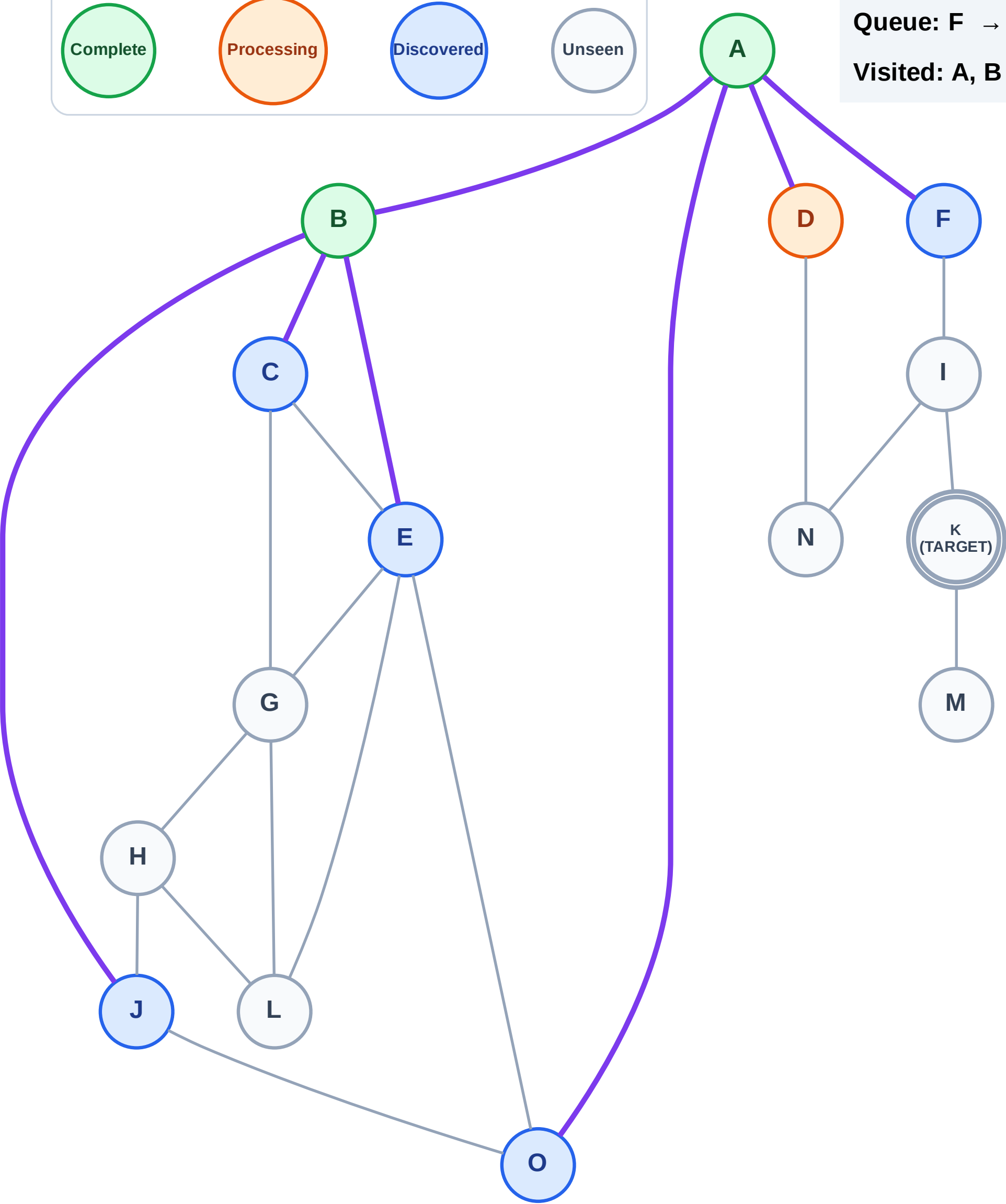
Processing

Discovered

Unseen

Queue: F → O → C → E → J

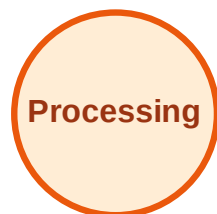
Visited: A, B



BFS Traversal

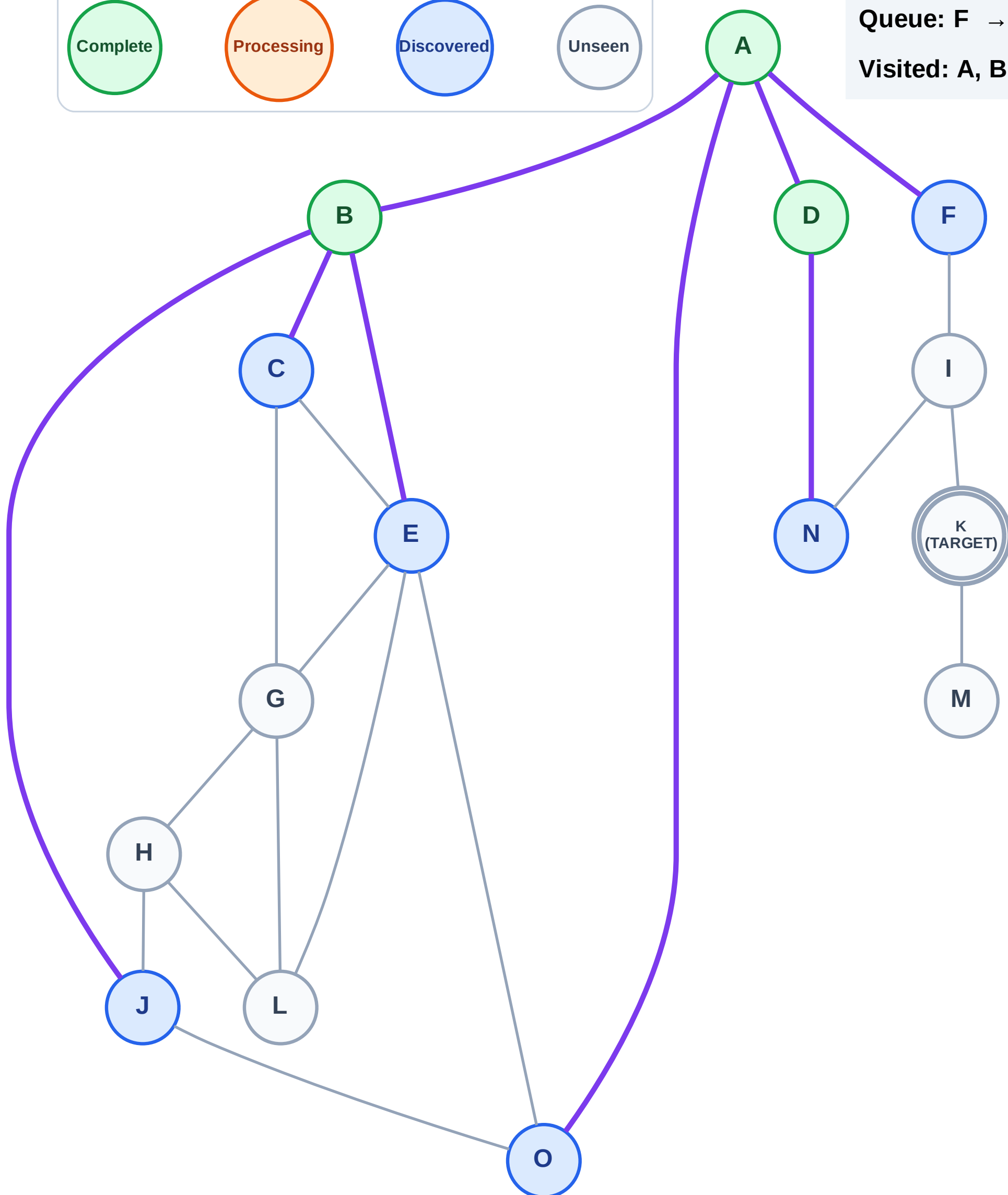
Step 7: Discover N from D

Legend



Queue: F → O → C → E → J → N

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

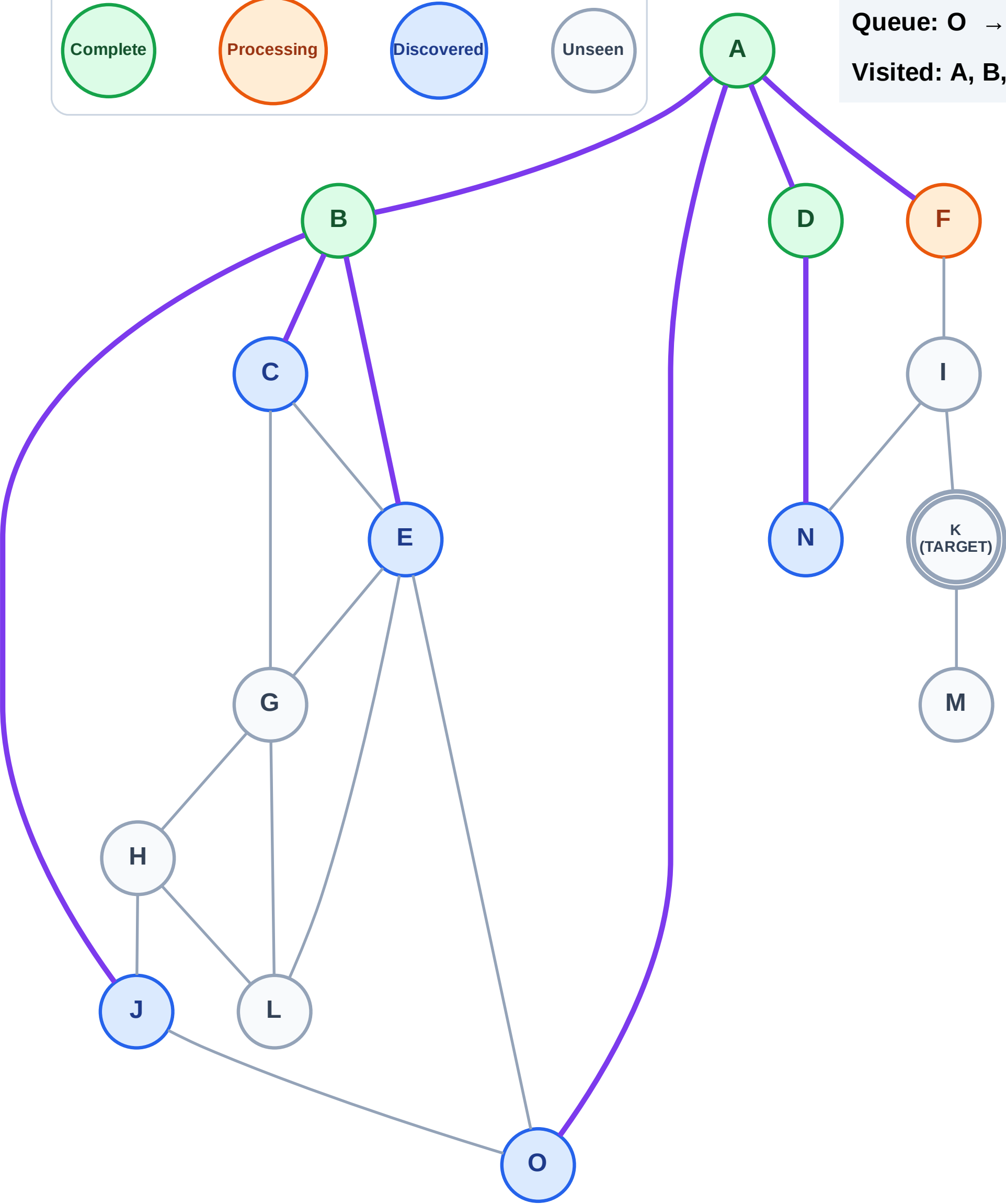
Processing

Discovered

Unseen

Queue: O → C → E → J → N

Visited: A, B, D



BFS Traversal

Step 9: Discover I from F

Legend

Complete

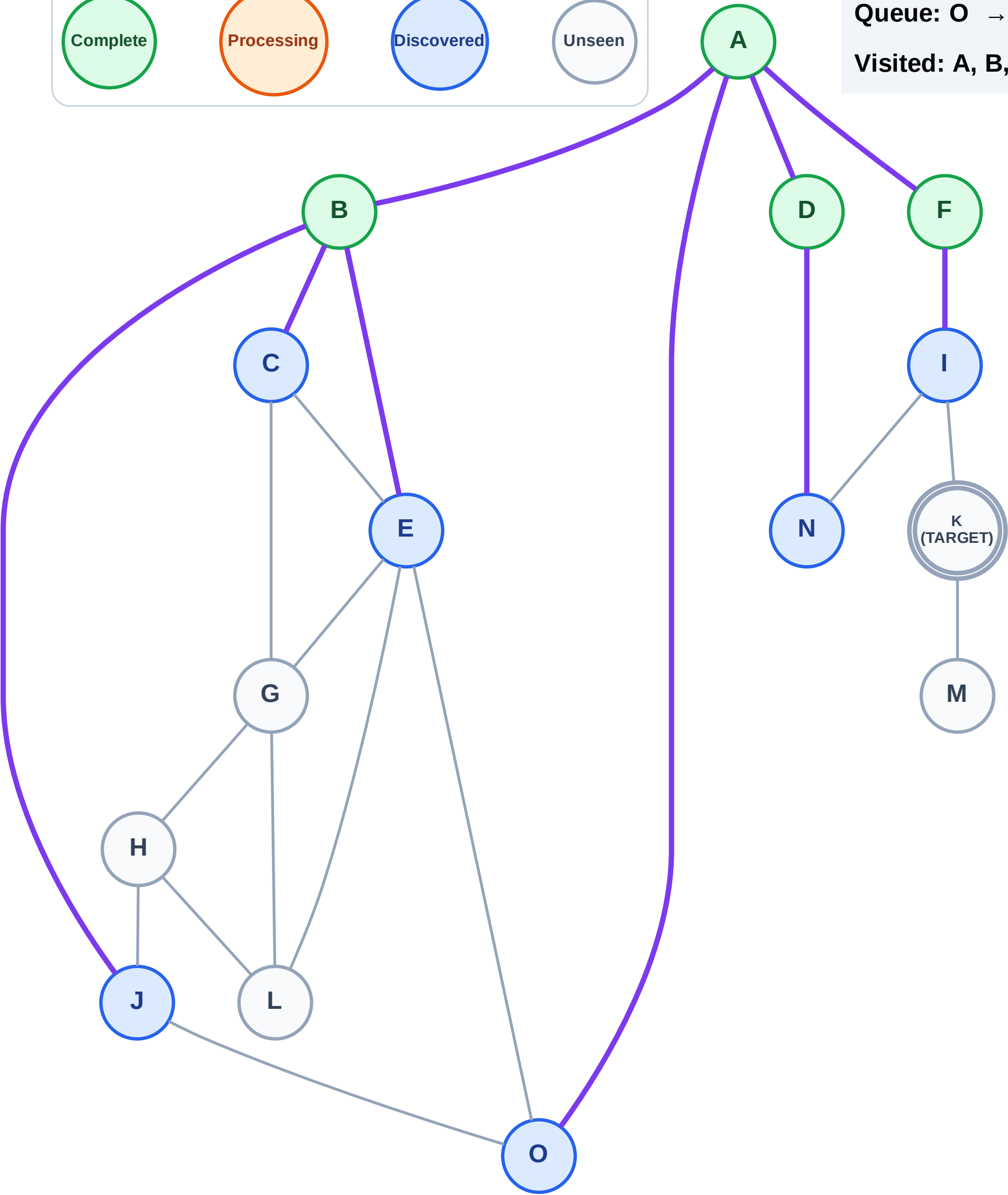
Processing

Discovered

Unseen

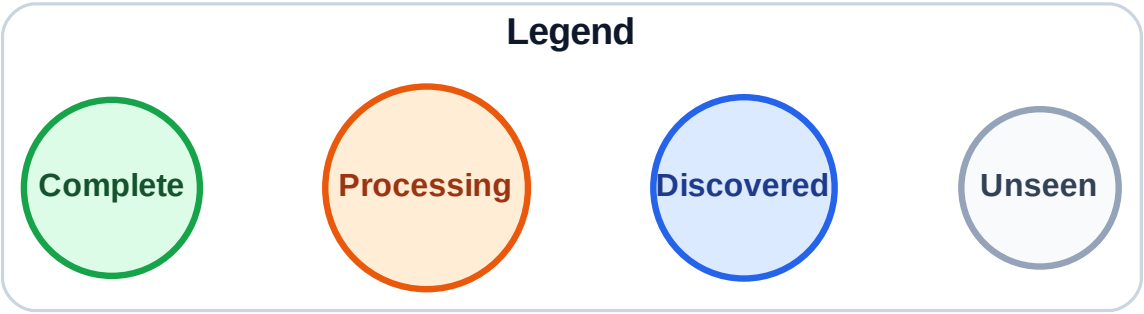
Queue: O → C → E → J → N → I

Visited: A, B, D, F



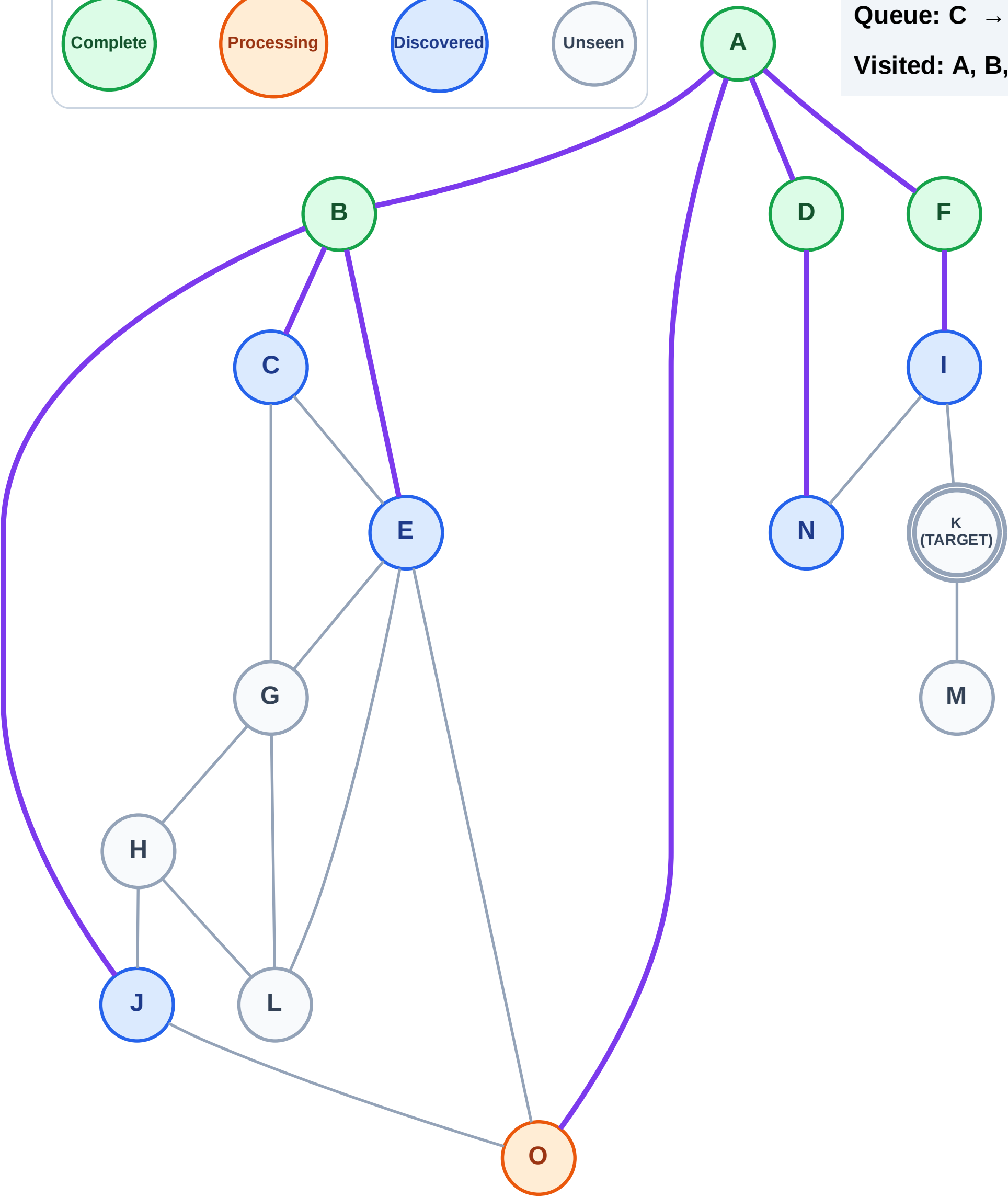
BFS Traversal

Step 10: Process node O



Queue: C → E → J → N → I

Visited: A, B, D, F



BFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



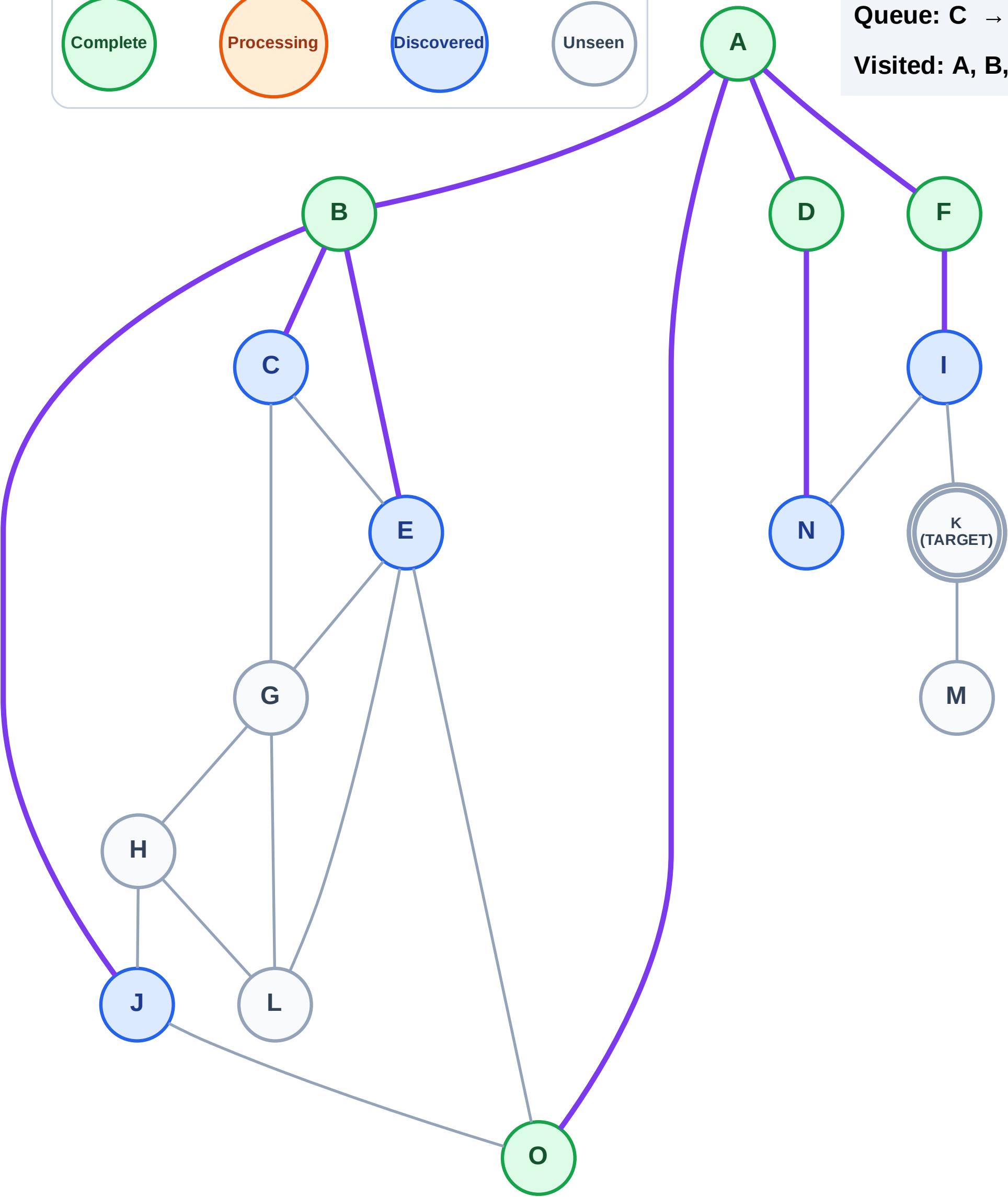
Discovered



Unseen

Queue: C → E → J → N → I

Visited: A, B, D, F, O



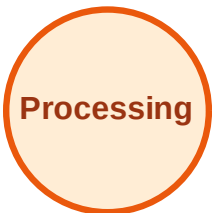
BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



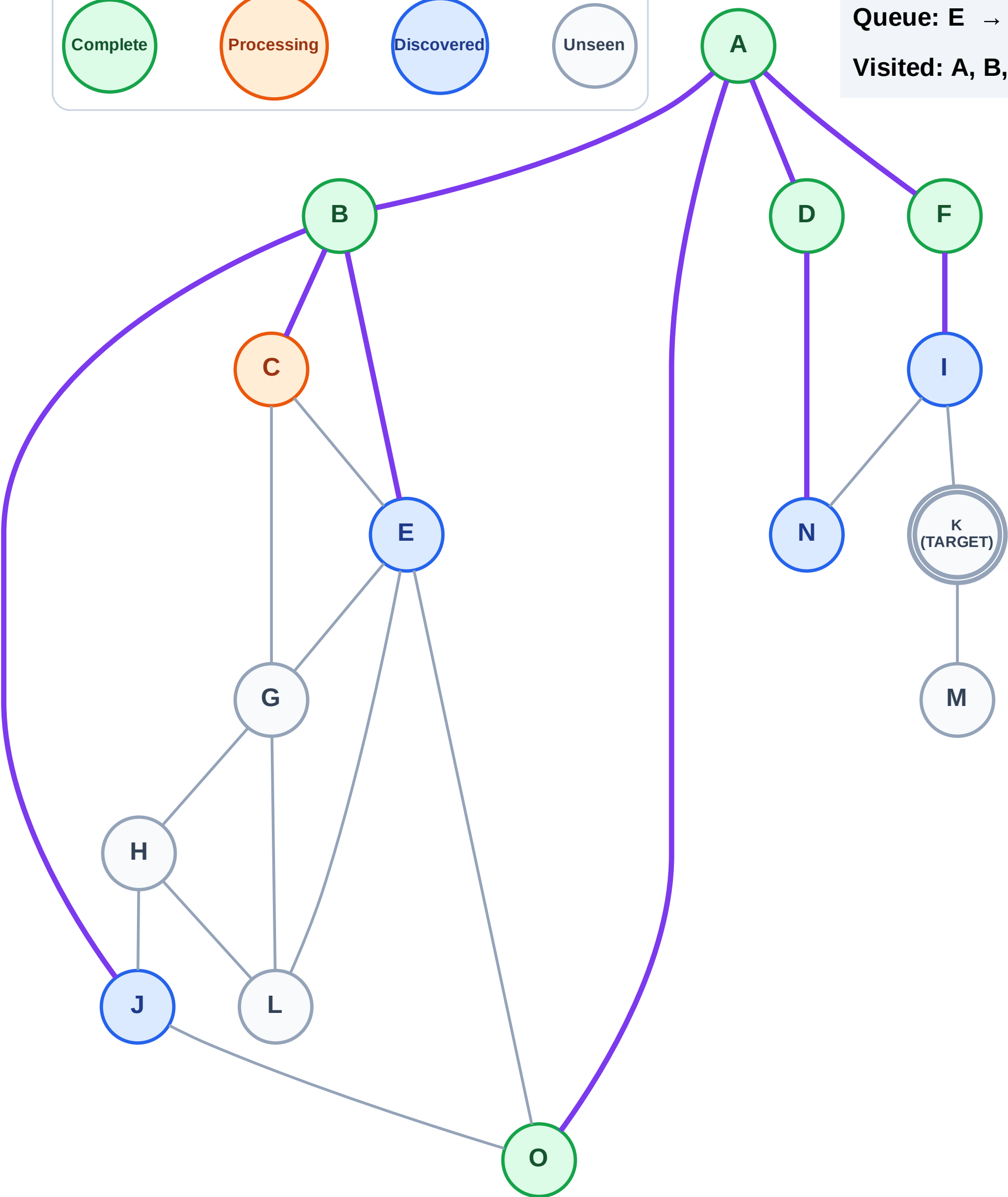
Discovered



Unseen

Queue: E → J → N → I

Visited: A, B, D, F, O



BFS Traversal

Step 13: Discover G from C

Legend



Complete



Processing



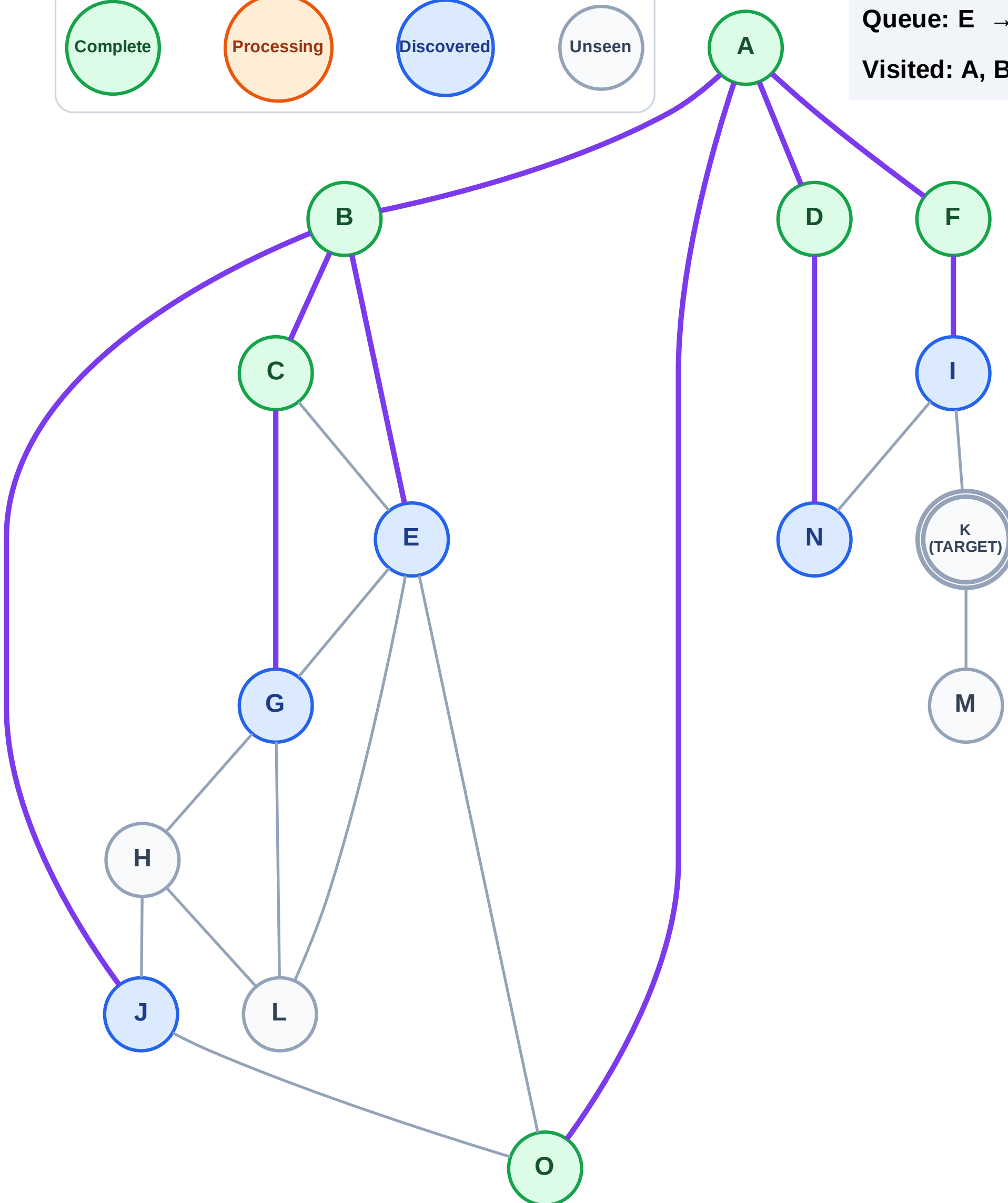
Discovered



Unseen

Queue: E → J → N → I → G

Visited: A, B, C, D, F, O



BFS Traversal

Step 14: Process node E

Legend

Complete

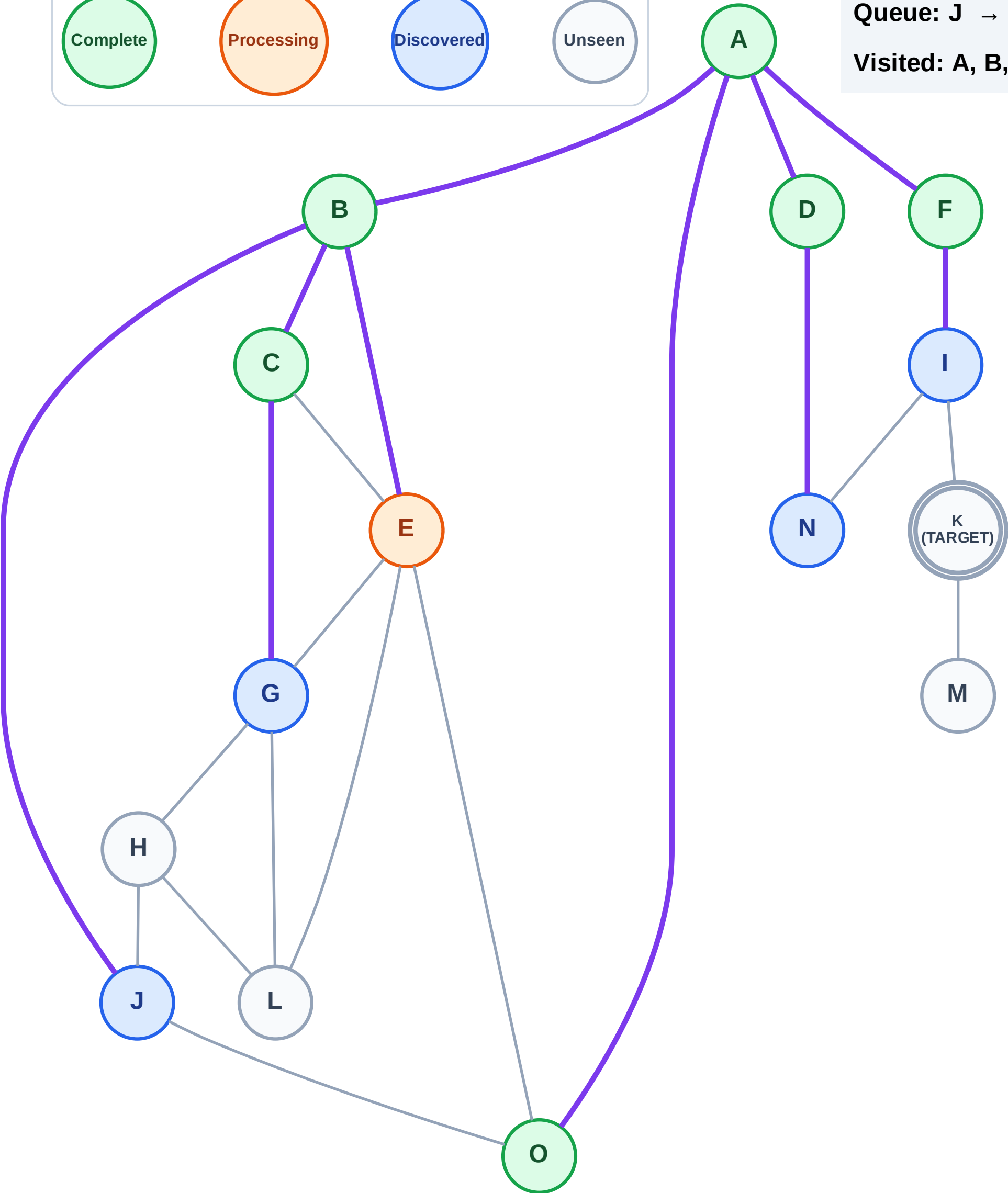
Processing

Discovered

Unseen

Queue: J → N → I → G

Visited: A, B, C, D, F, O



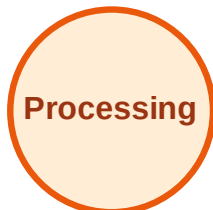
BFS Traversal

Step 15: Discover L from E

Legend



Complete



Processing



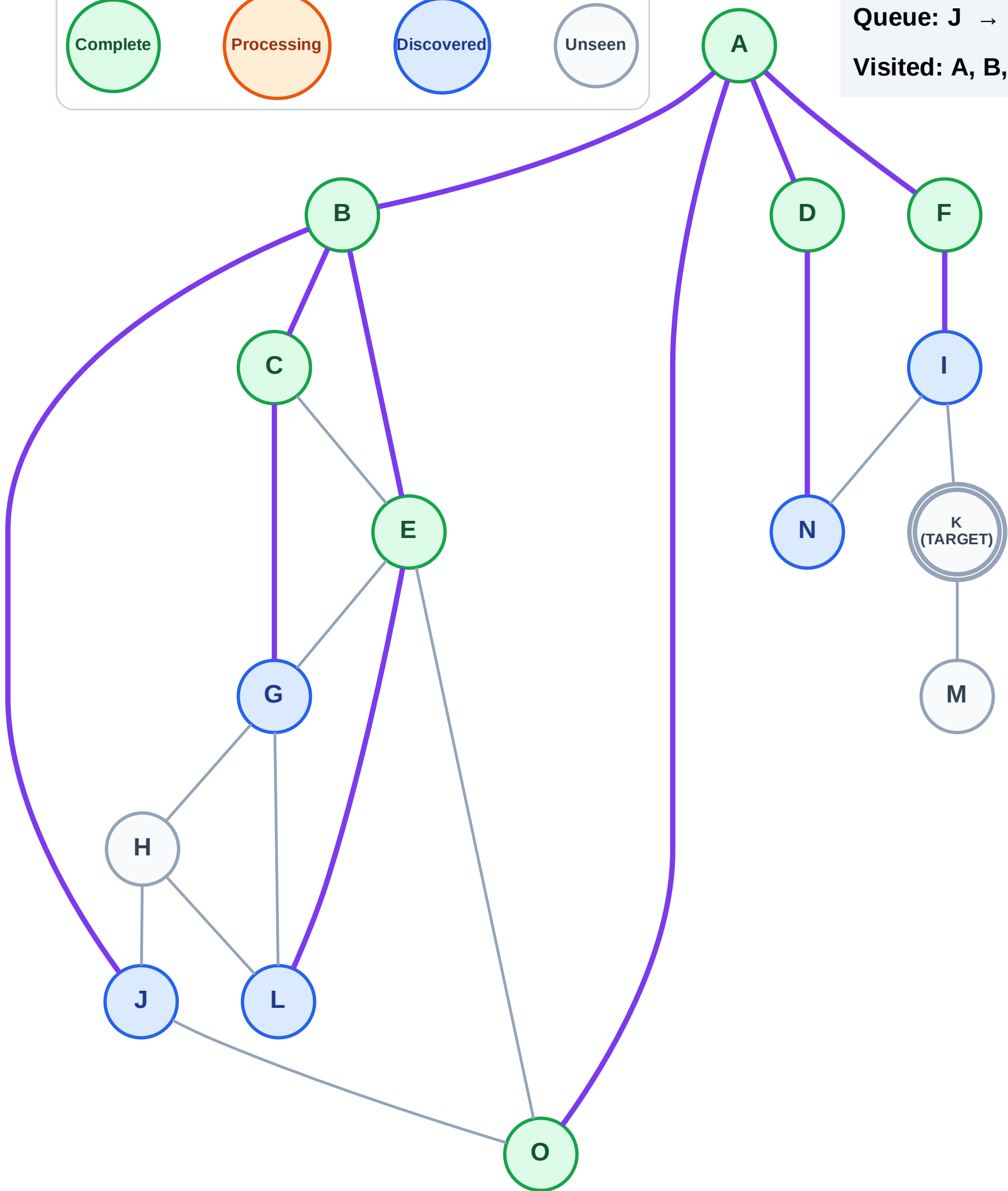
Discovered



Unseen

Queue: J → N → I → G → L

Visited: A, B, C, D, E, F, O



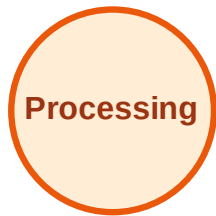
BFS Traversal

Step 16: Process node J

Legend



Complete



Processing



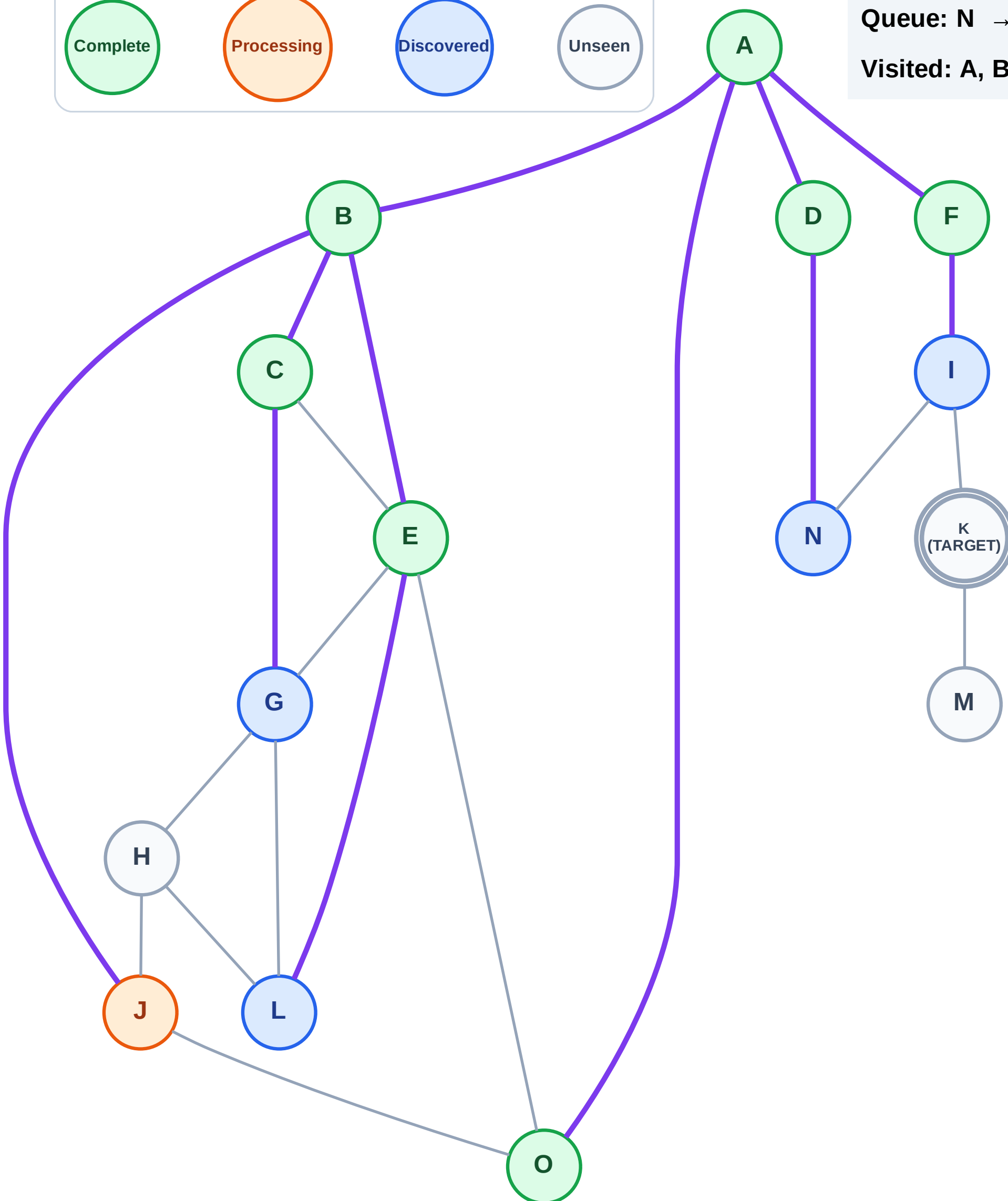
Discovered



Unseen

Queue: N → I → G → L

Visited: A, B, C, D, E, F, O



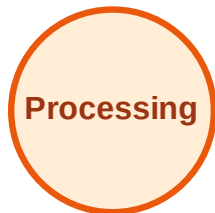
BFS Traversal

Step 17: Discover H from J

Legend



Complete



Processing



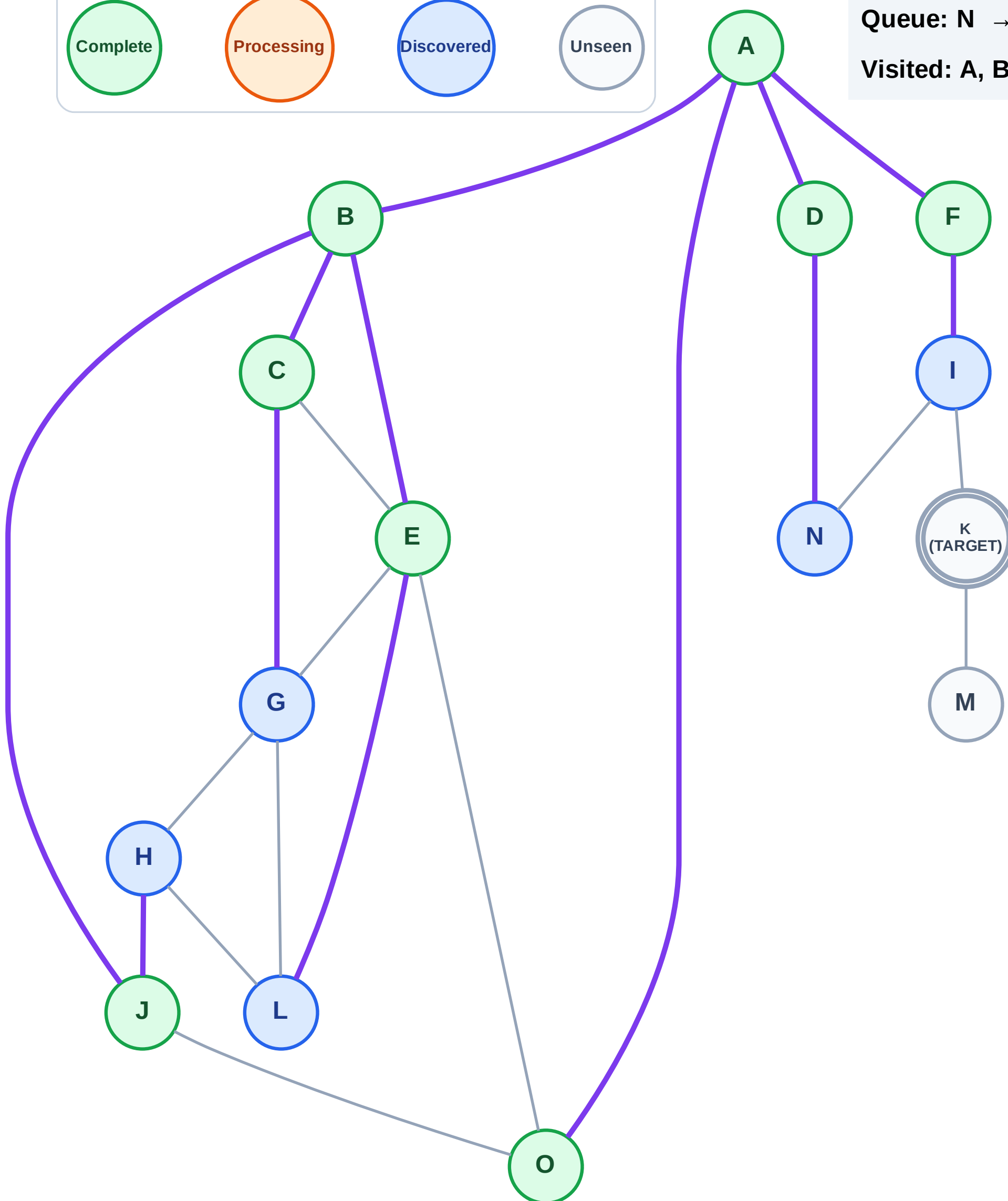
Discovered



Unseen

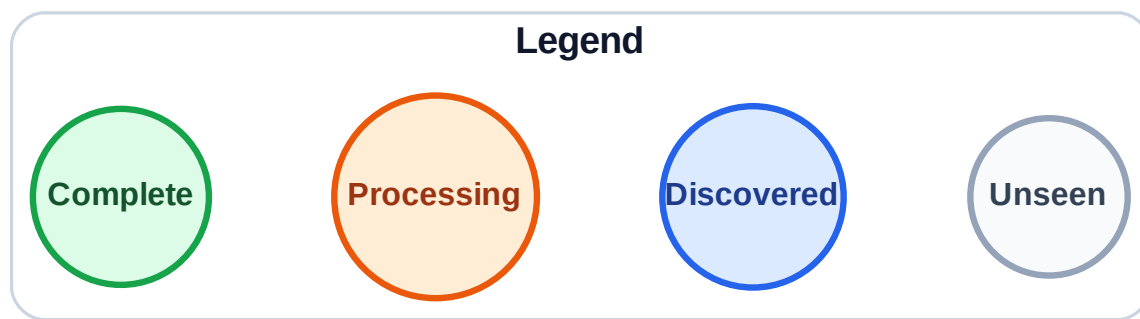
Queue: N → I → G → L → H

Visited: A, B, C, D, E, F, J, O

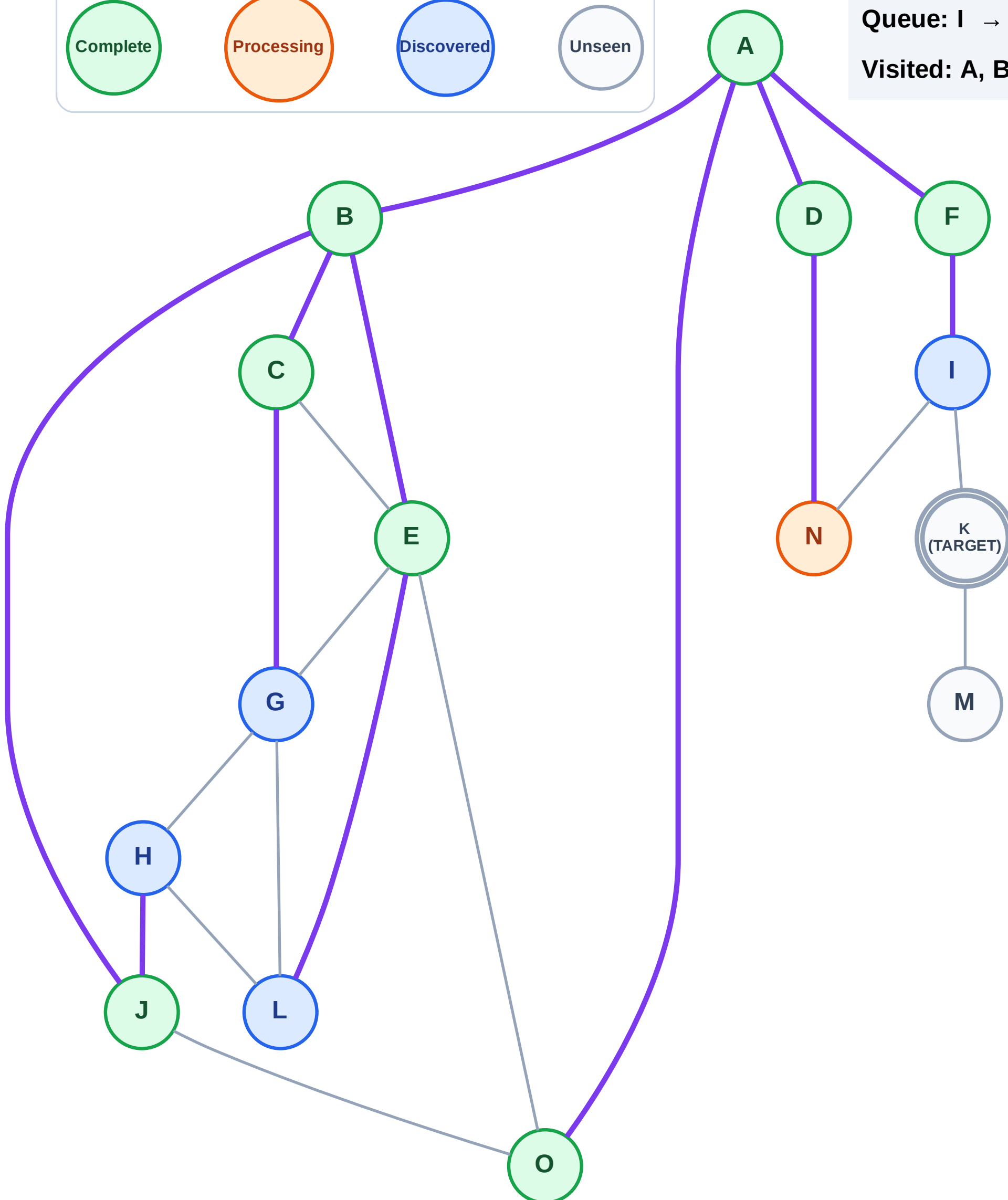


BFS Traversal

Step 18: Process node N



Queue: I → G → L → H
Visited: A, B, C, D, E, F, J, O



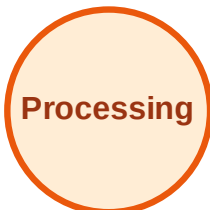
BFS Traversal

Step 19: No new neighbors from N

Legend



Complete



Processing



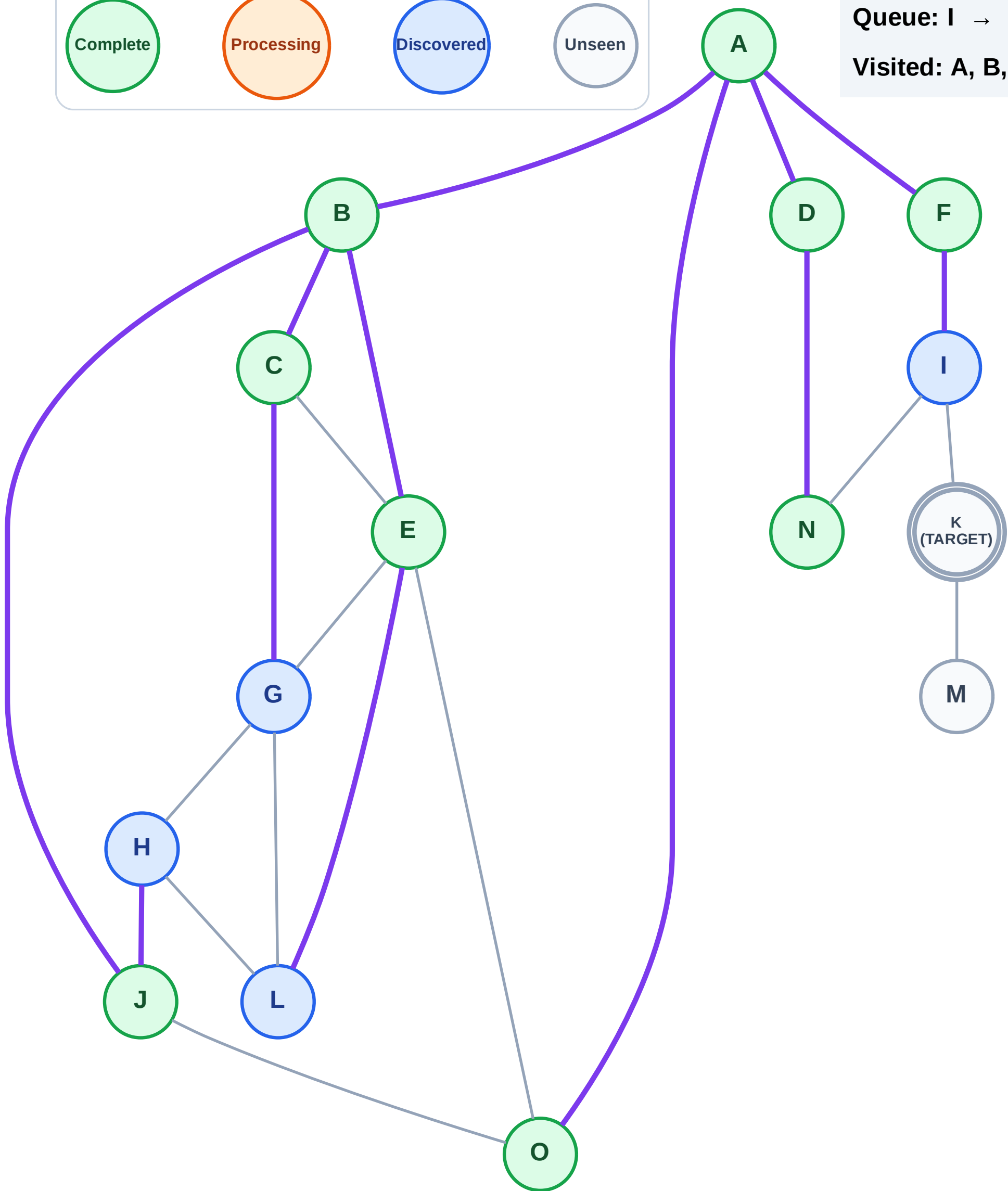
Discovered



Unseen

Queue: I → G → L → H

Visited: A, B, C, D, E, F, J, N, O



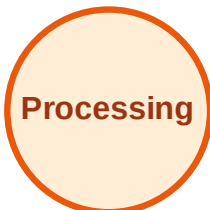
BFS Traversal

Step 20: Process node I

Legend



Complete



Processing



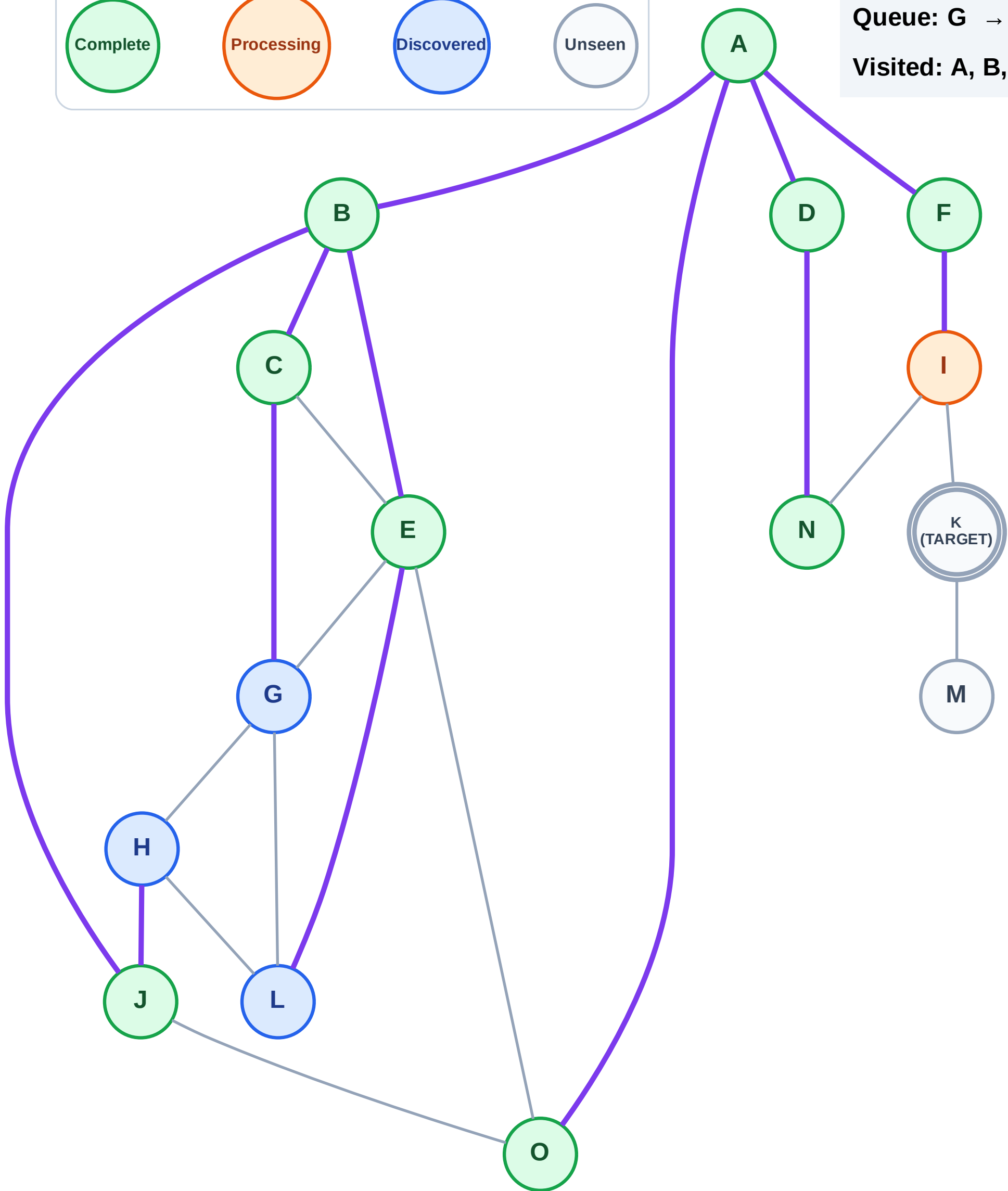
Discovered



Unseen

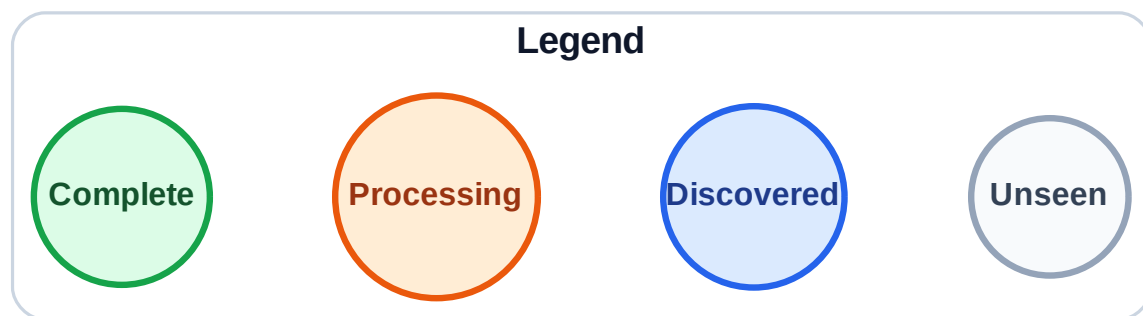
Queue: G → L → H

Visited: A, B, C, D, E, F, J, N, O

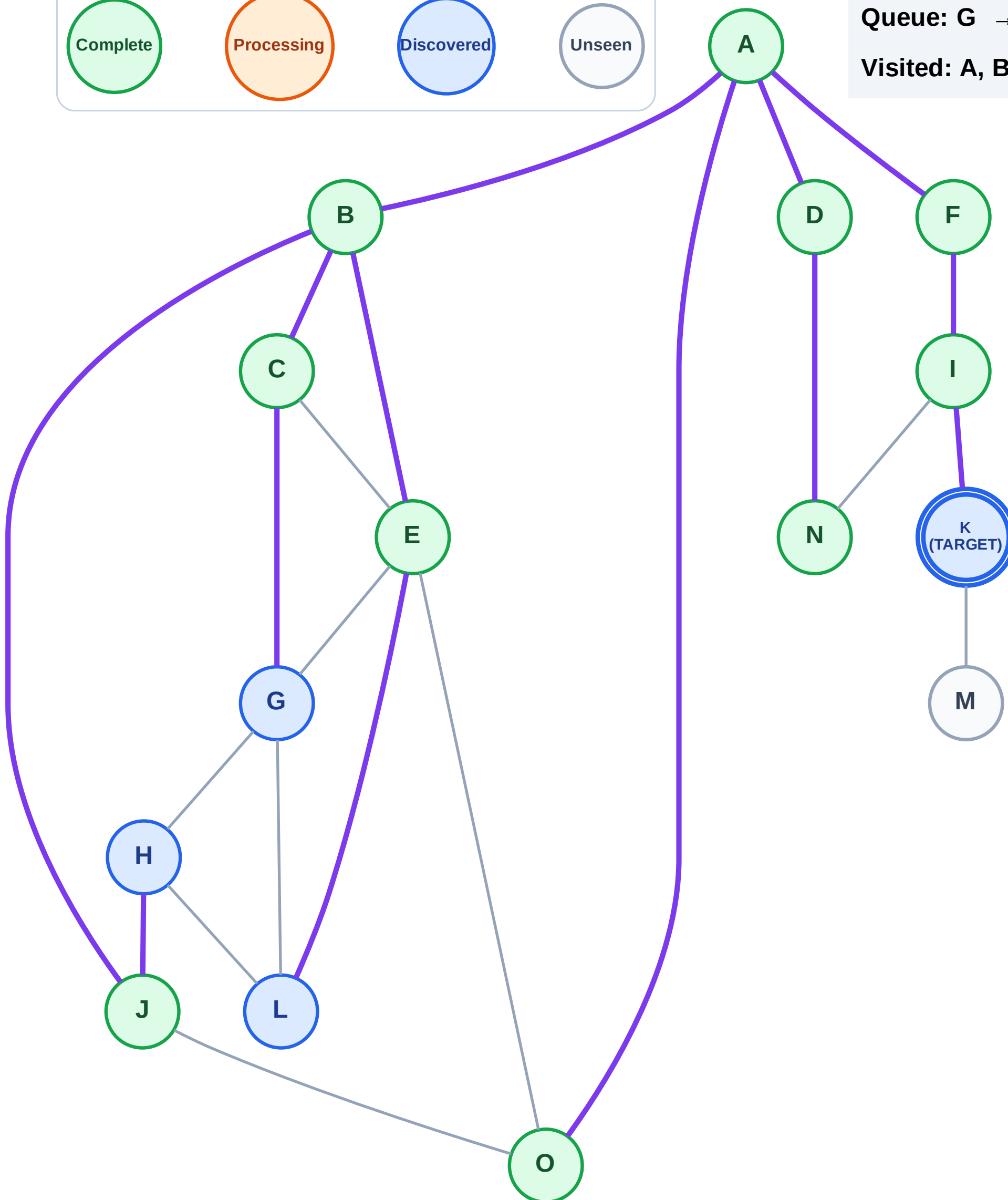


BFS Traversal

Step 21: Discover K from I



Queue: G → L → H → K
Visited: A, B, C, D, E, F, I, J, N, O



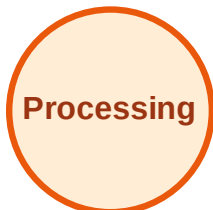
BFS Traversal

Step 22: Process node G

Legend



Complete



Processing



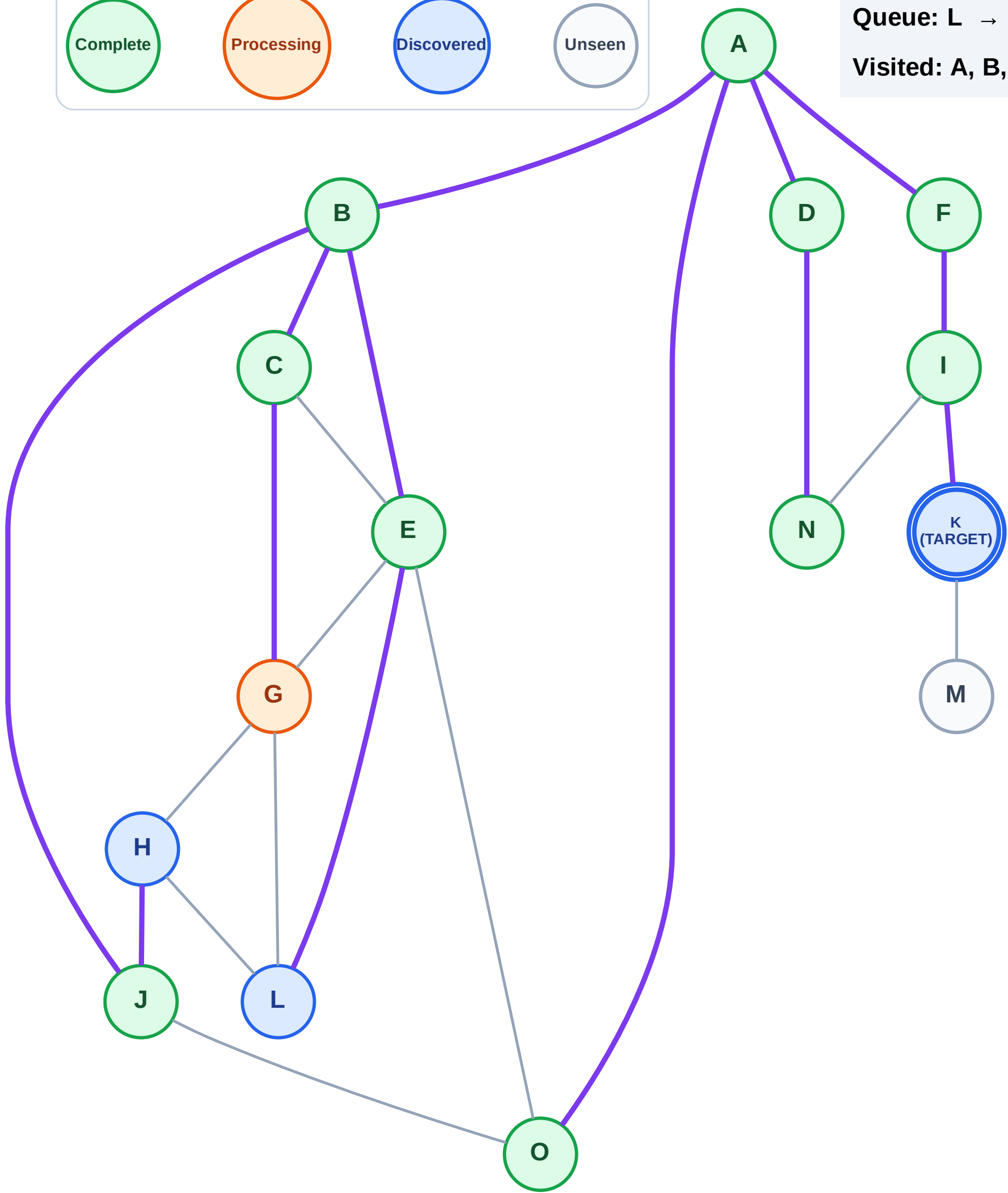
Discovered



Unseen

Queue: L → H → K

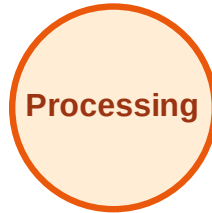
Visited: A, B, C, D, E, F, I, J, N, O



BFS Traversal

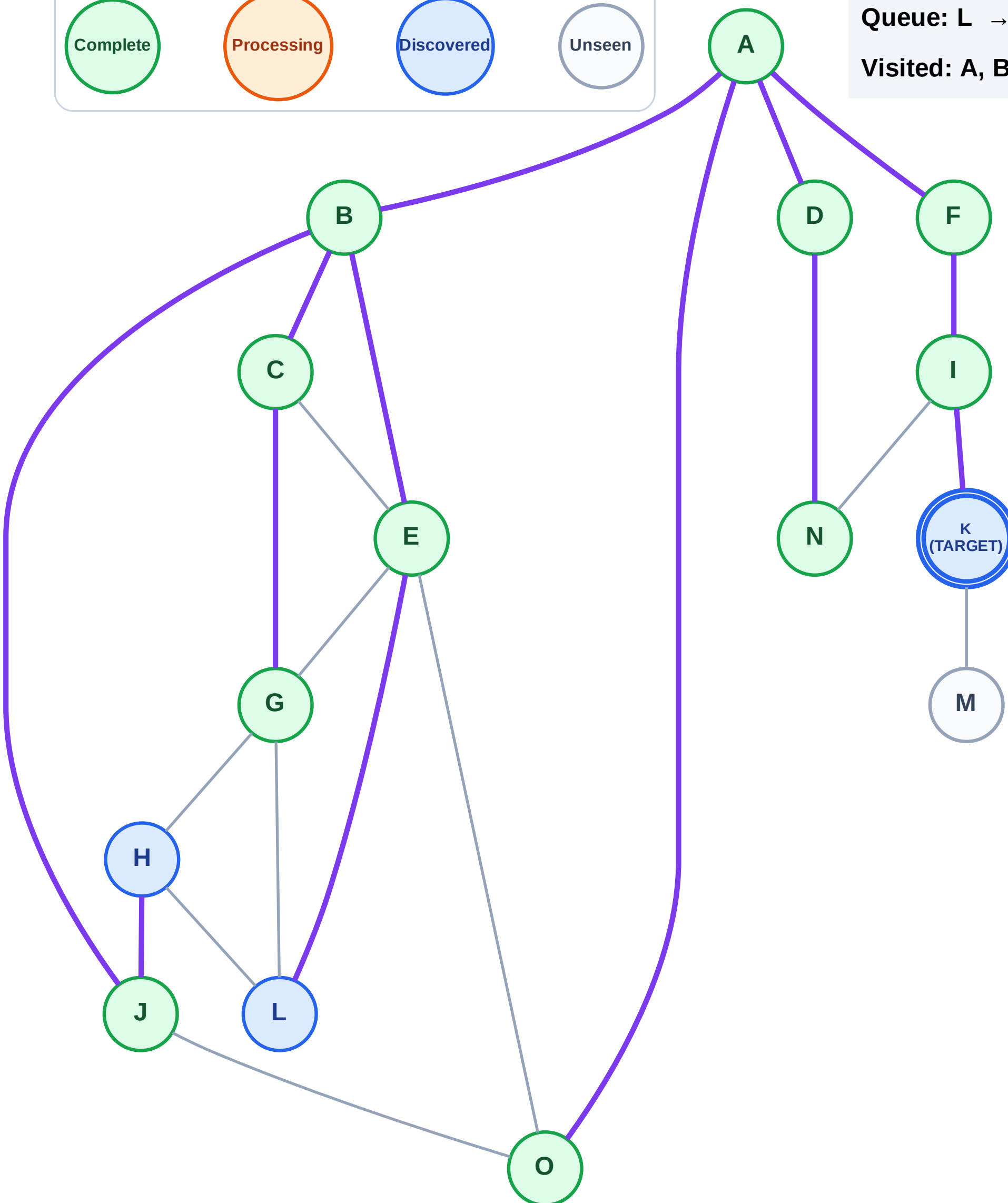
Step 23: No new neighbors from G

Legend



Queue: L → H → K

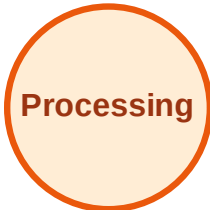
Visited: A, B, C, D, E, F, G, I, J, N, O



BFS Traversal

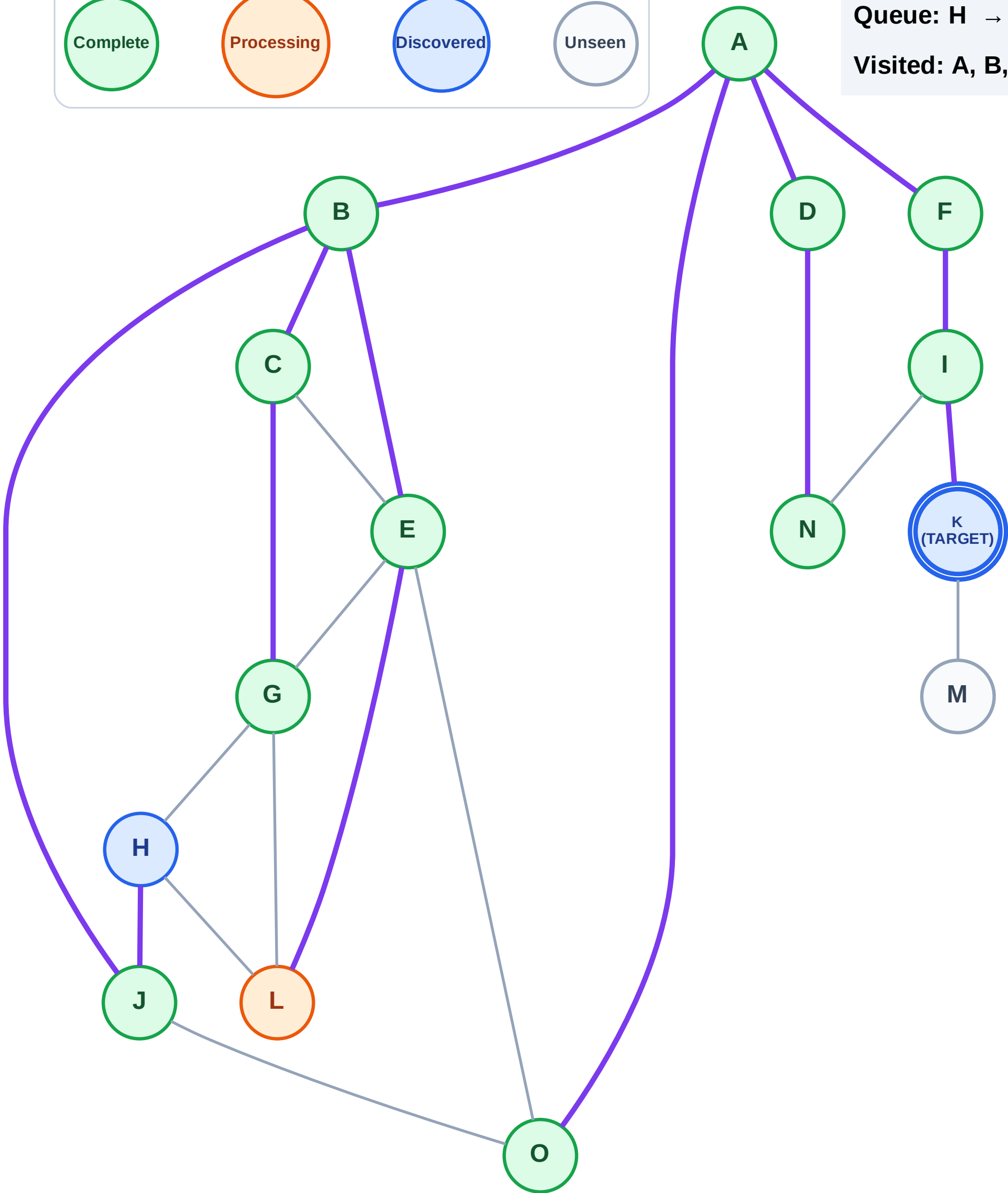
Step 24: Process node L

Legend



Queue: H → K

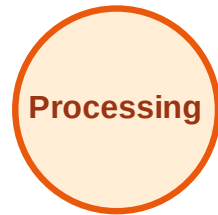
Visited: A, B, C, D, E, F, G, I, J, N, O



BFS Traversal

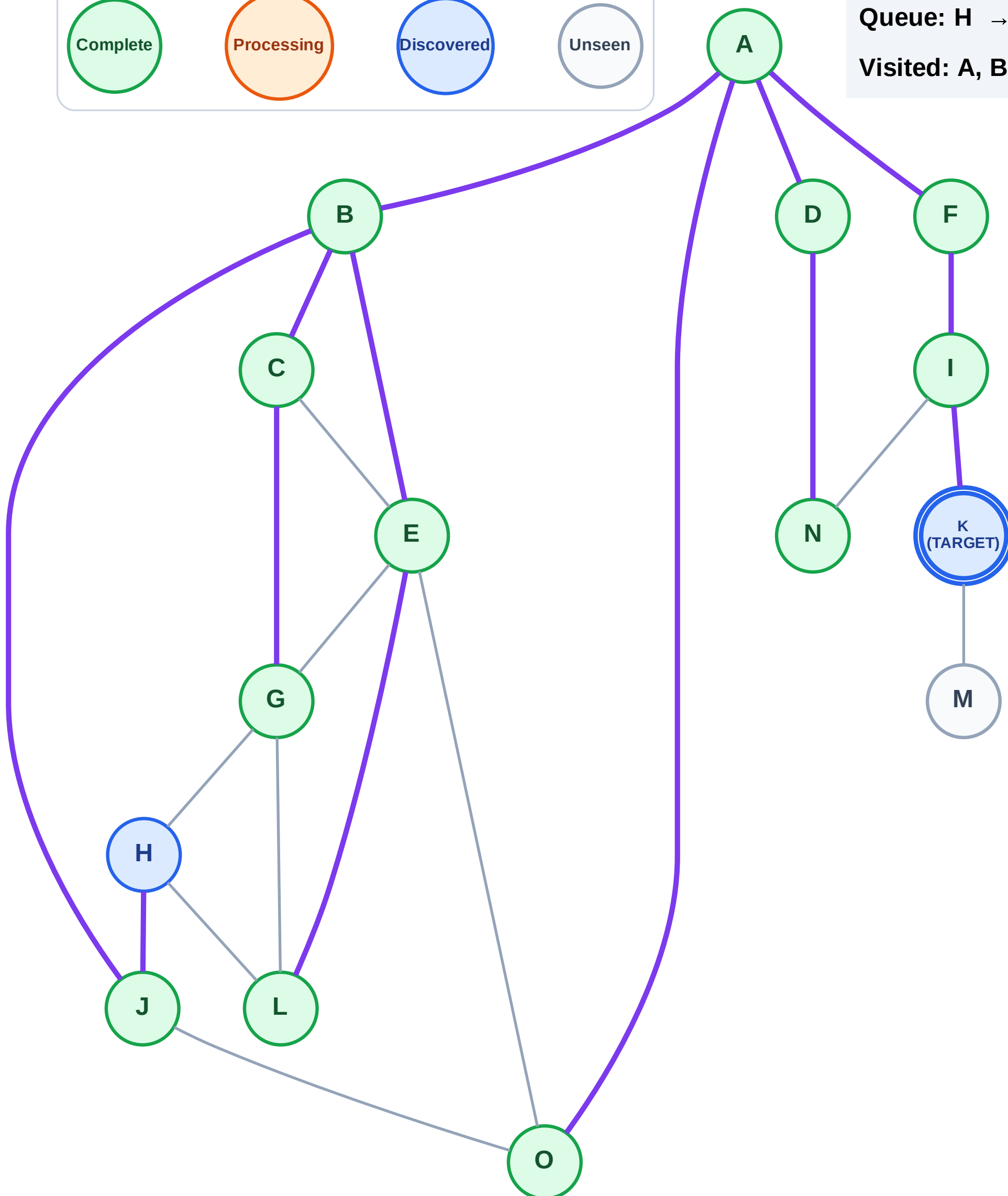
Step 25: No new neighbors from L

Legend



Queue: H → K

Visited: A, B, C, D, E, F, G, I, J, L, N, O



BFS Traversal

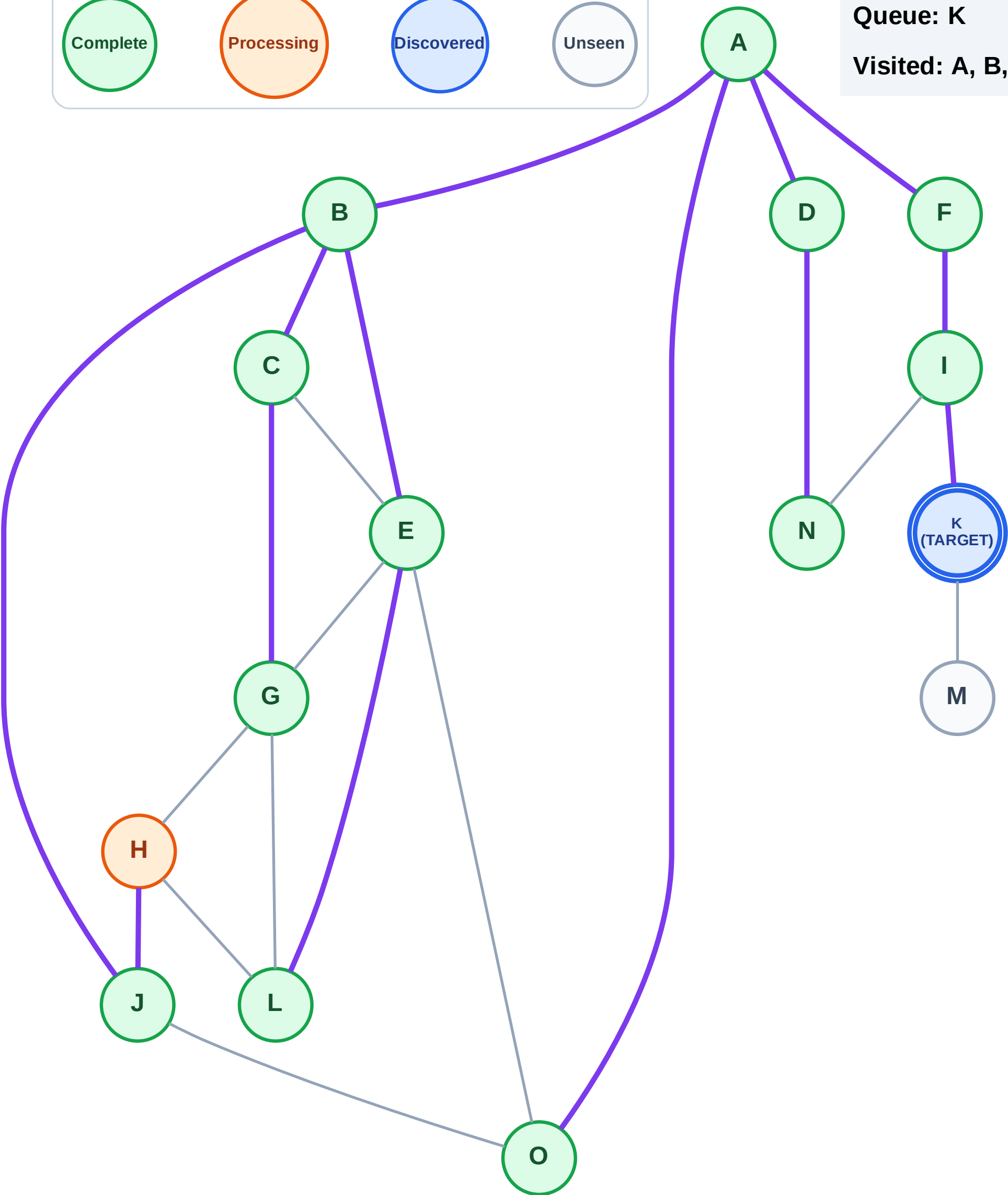
Step 26: Process node H

Legend



Queue: K

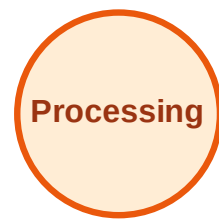
Visited: A, B, C, D, E, F, G, I, J, L, N, O



BFS Traversal

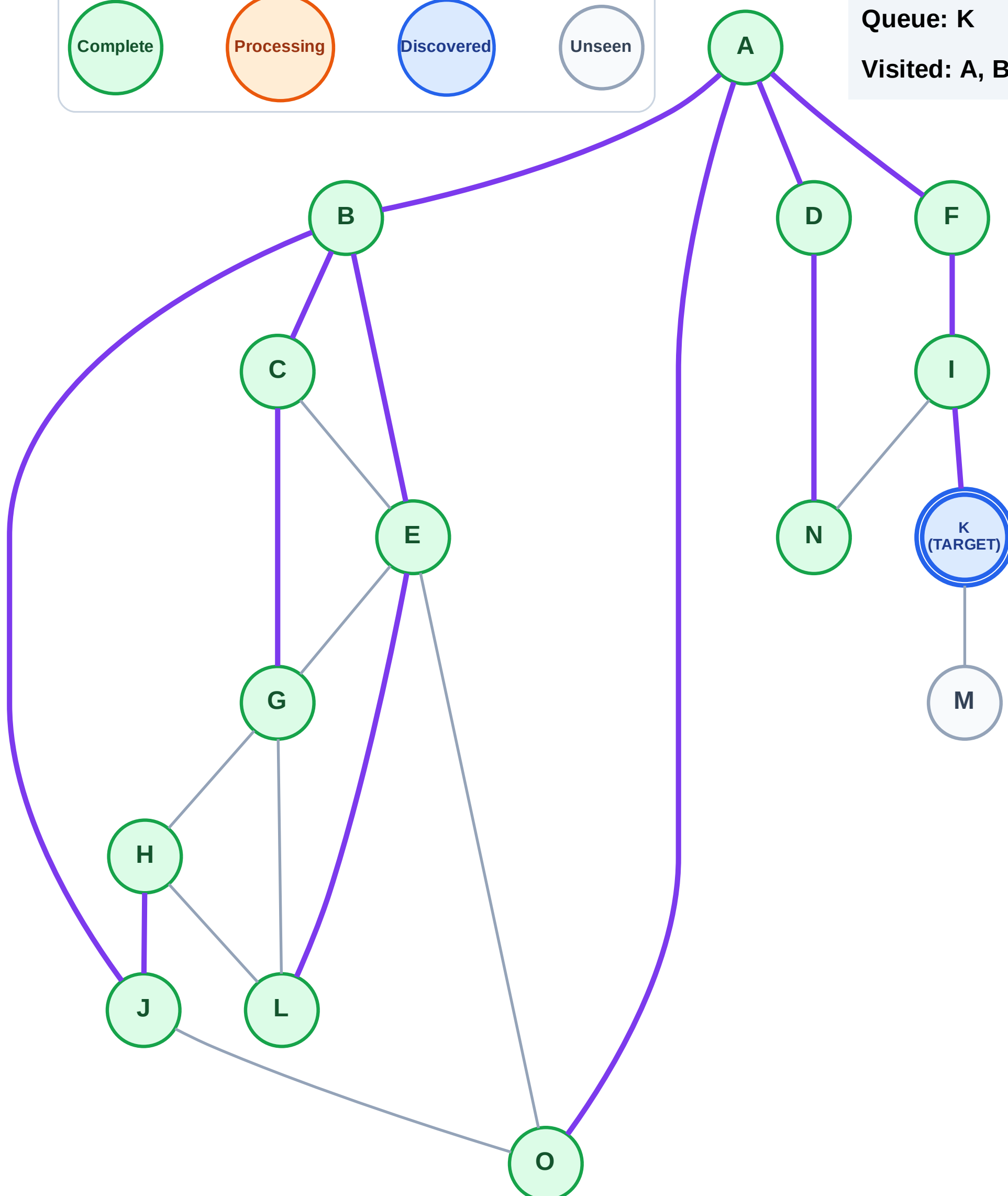
Step 27: No new neighbors from H

Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



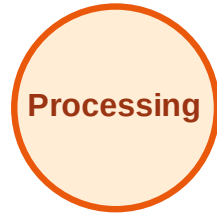
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



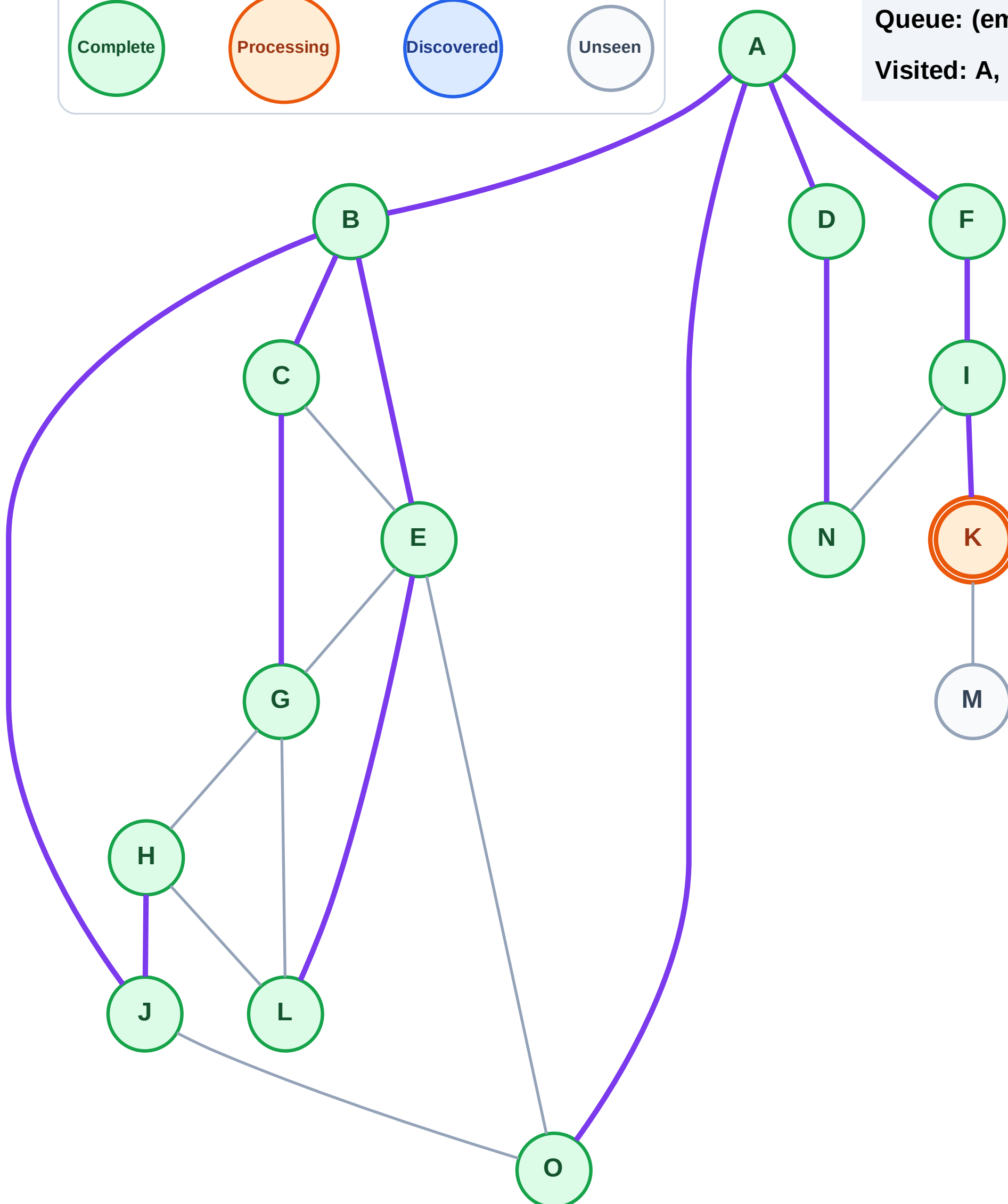
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → F → I → K

Legend

Complete

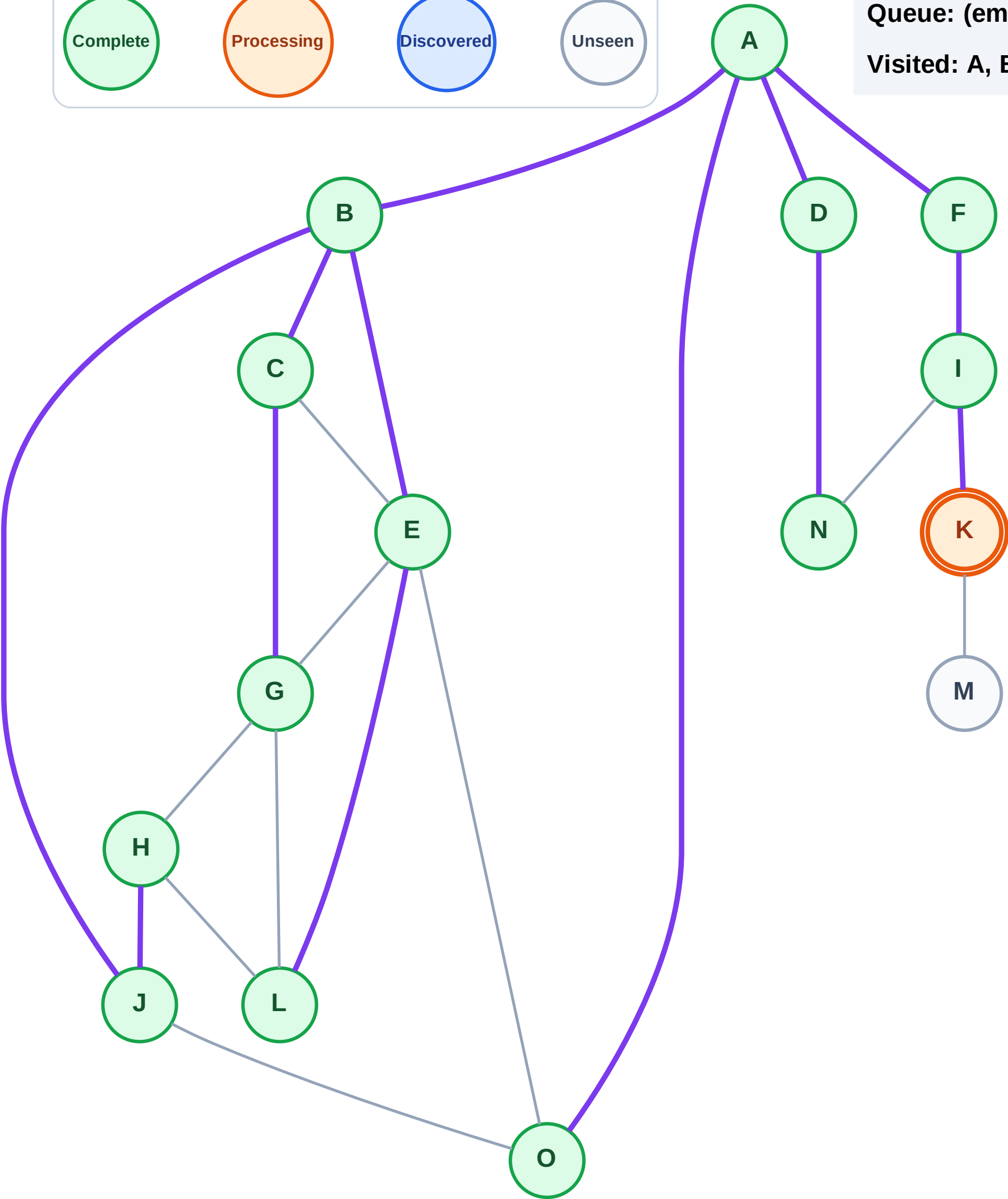
Processing

Discovered

Unseen

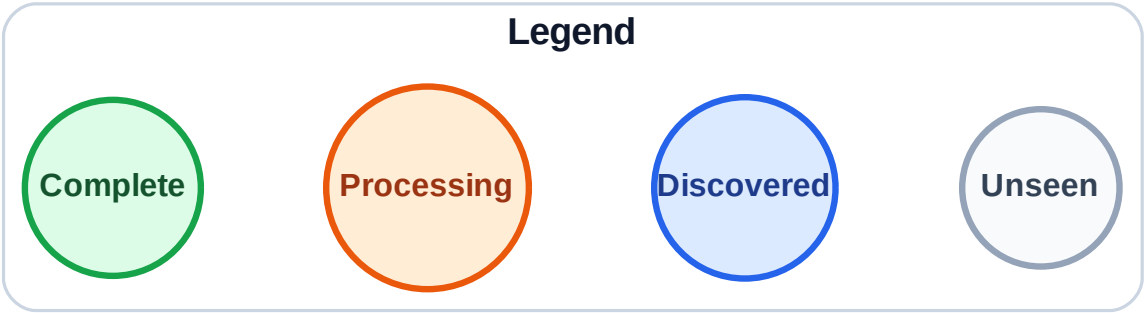
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, N, O

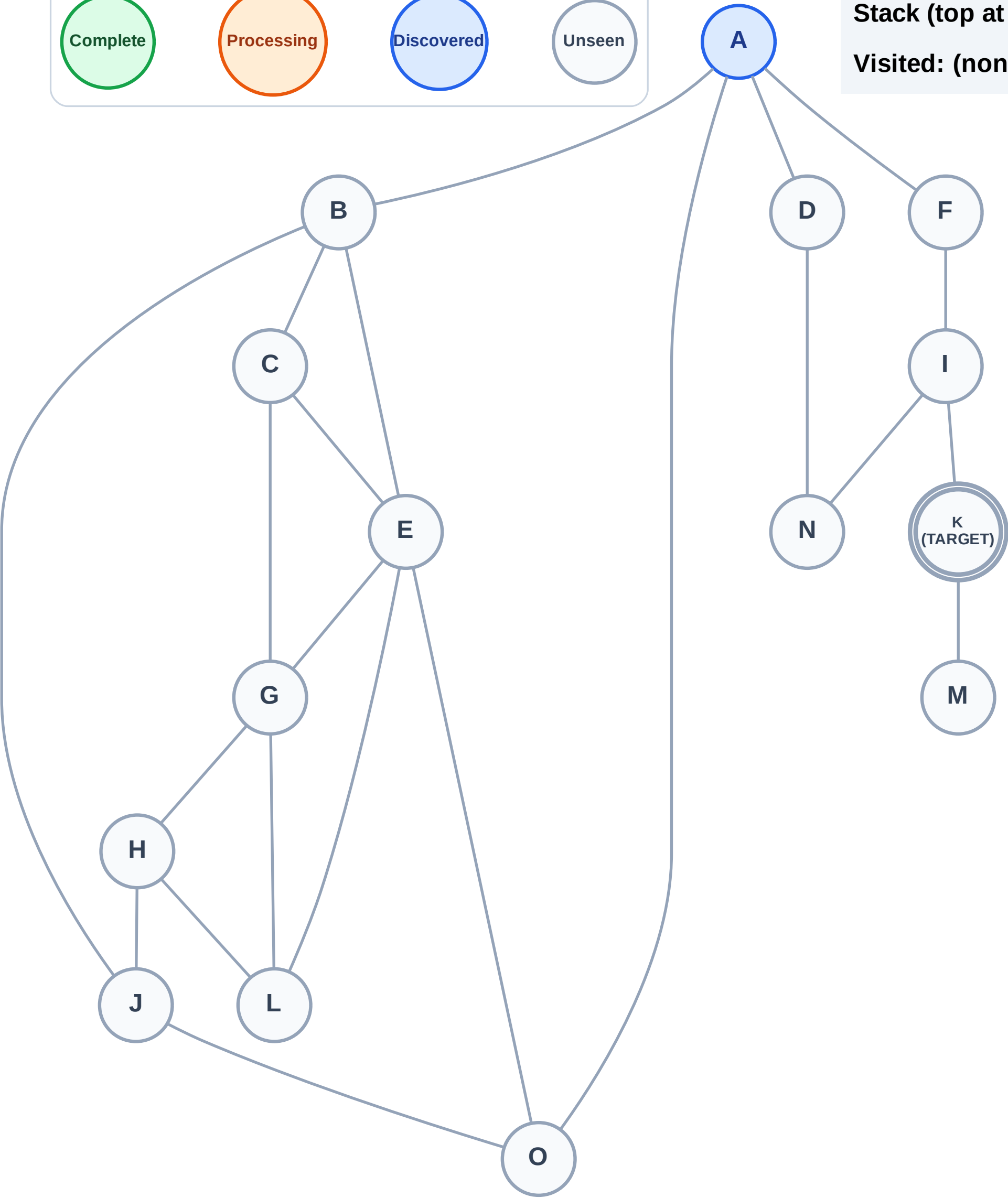


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

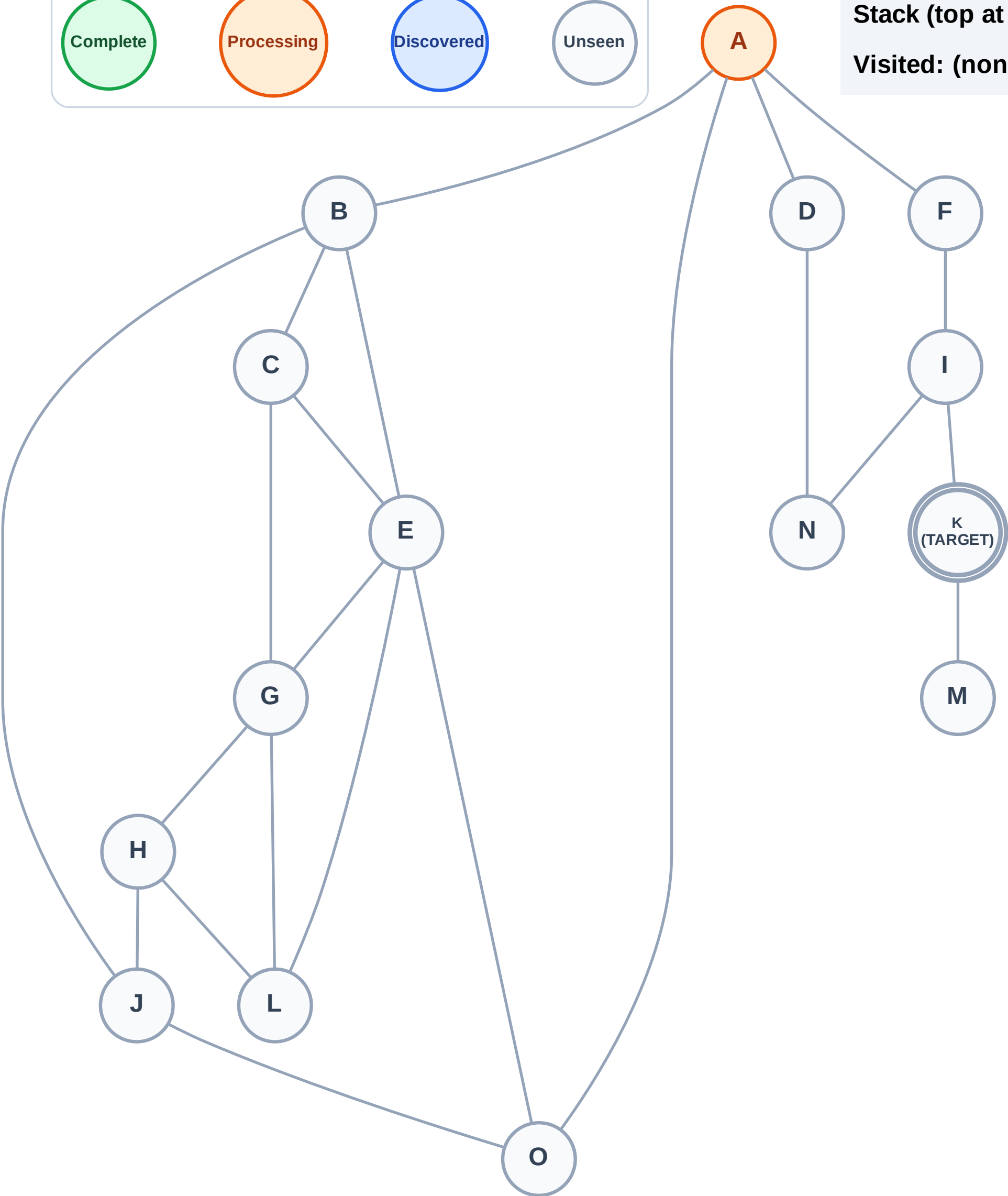
Complete

Processing

Discovered

Unseen

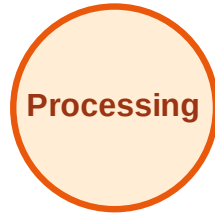
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

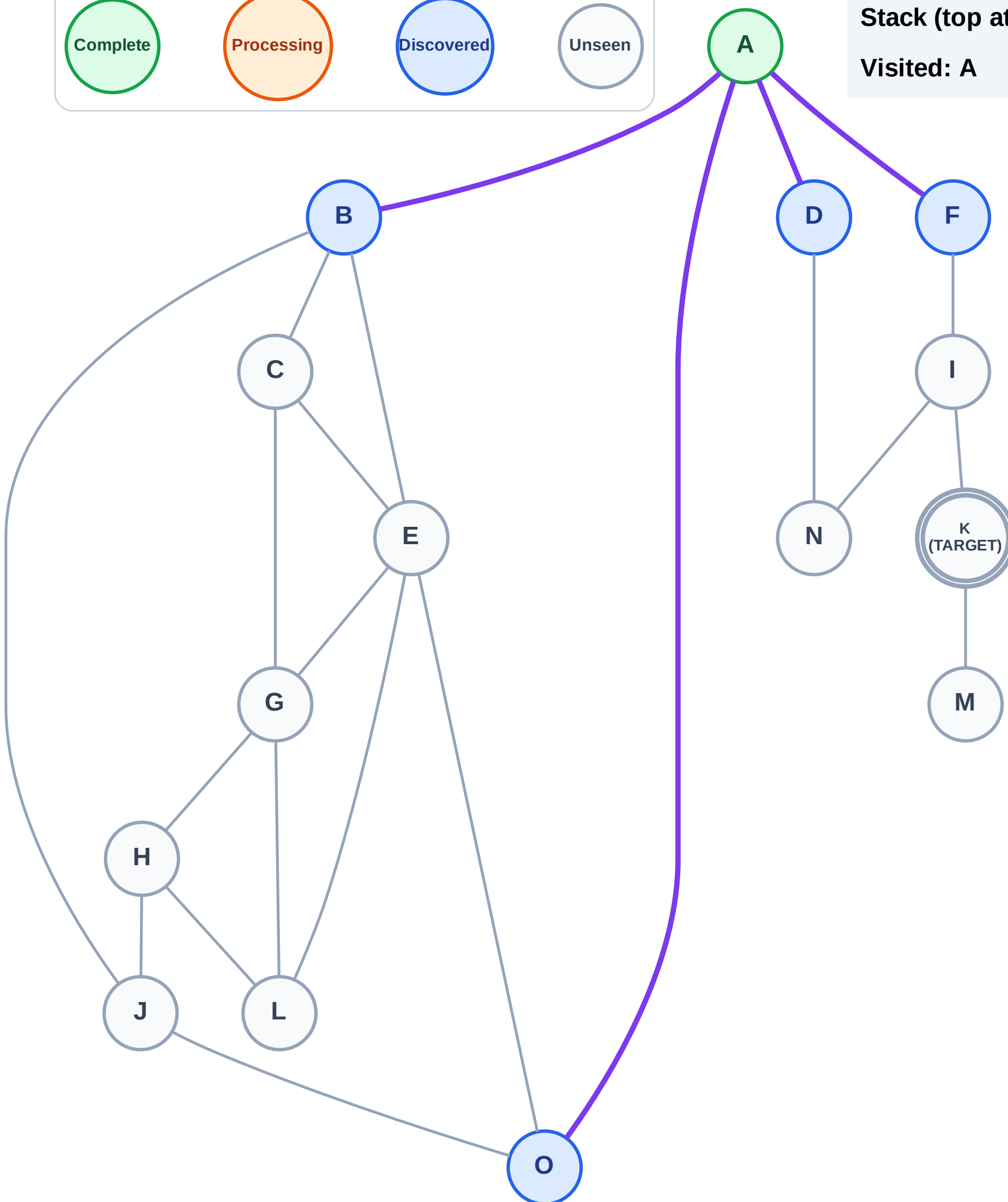
Step 3: Discover O, F, D, B from A

Legend



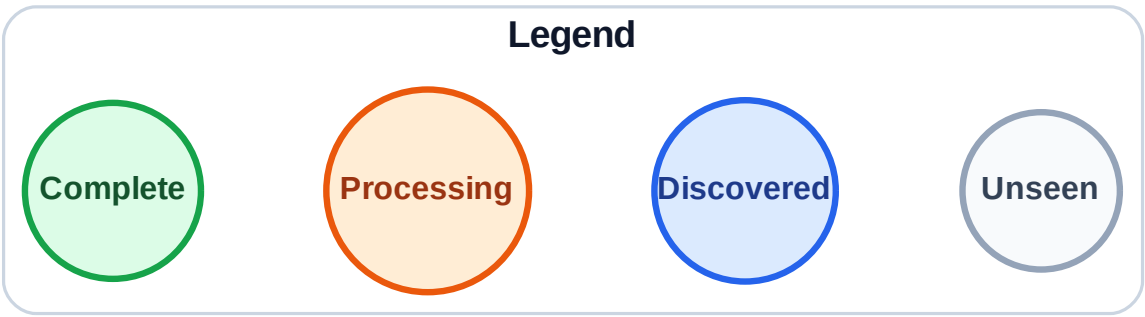
Stack (top at right): O | F | D | B

Visited: A



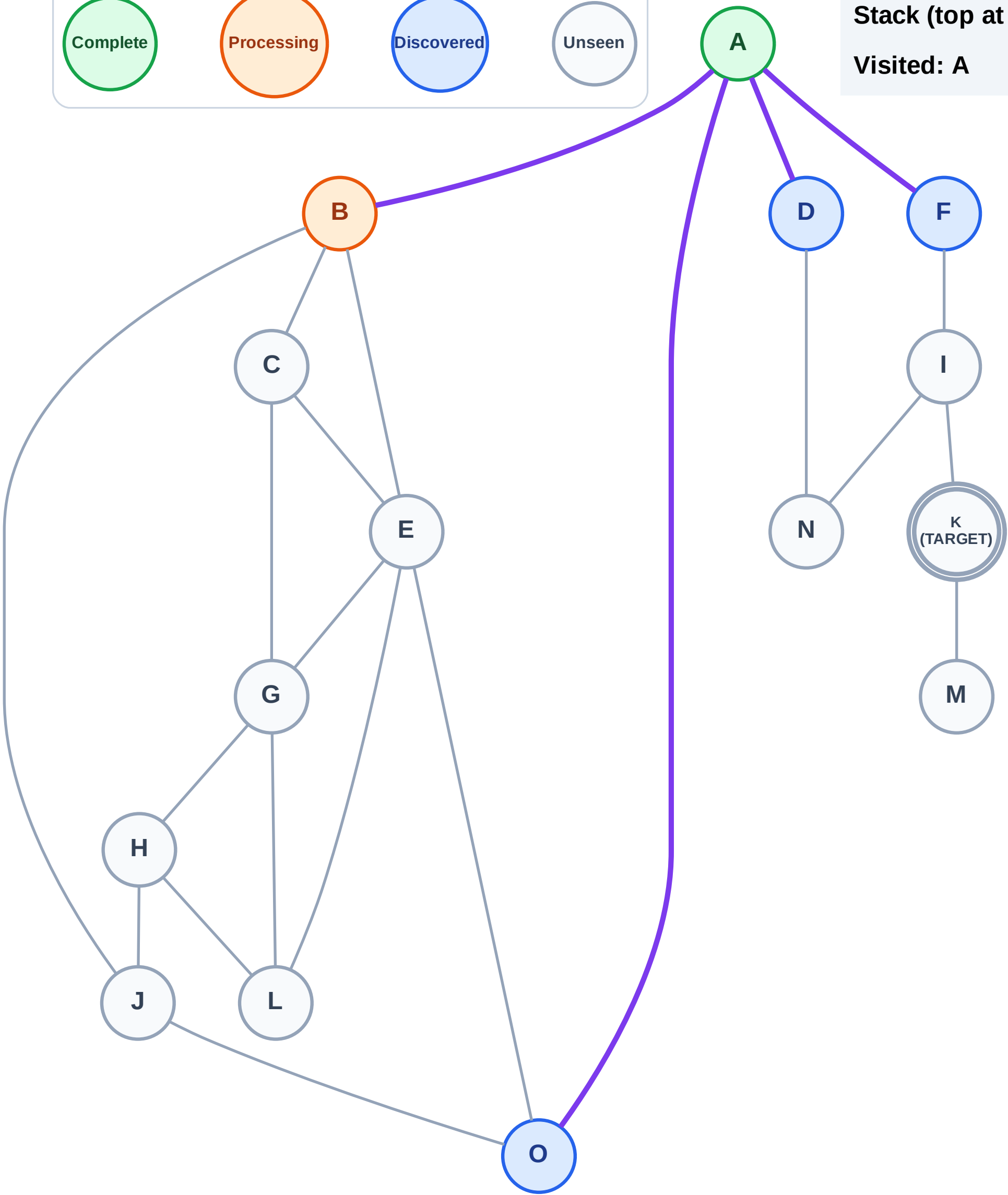
DFS Traversal

Step 4: Process node B



Stack (top at right): O | F | D

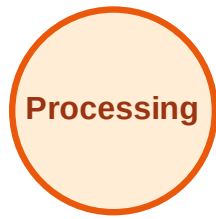
Visited: A



DFS Traversal

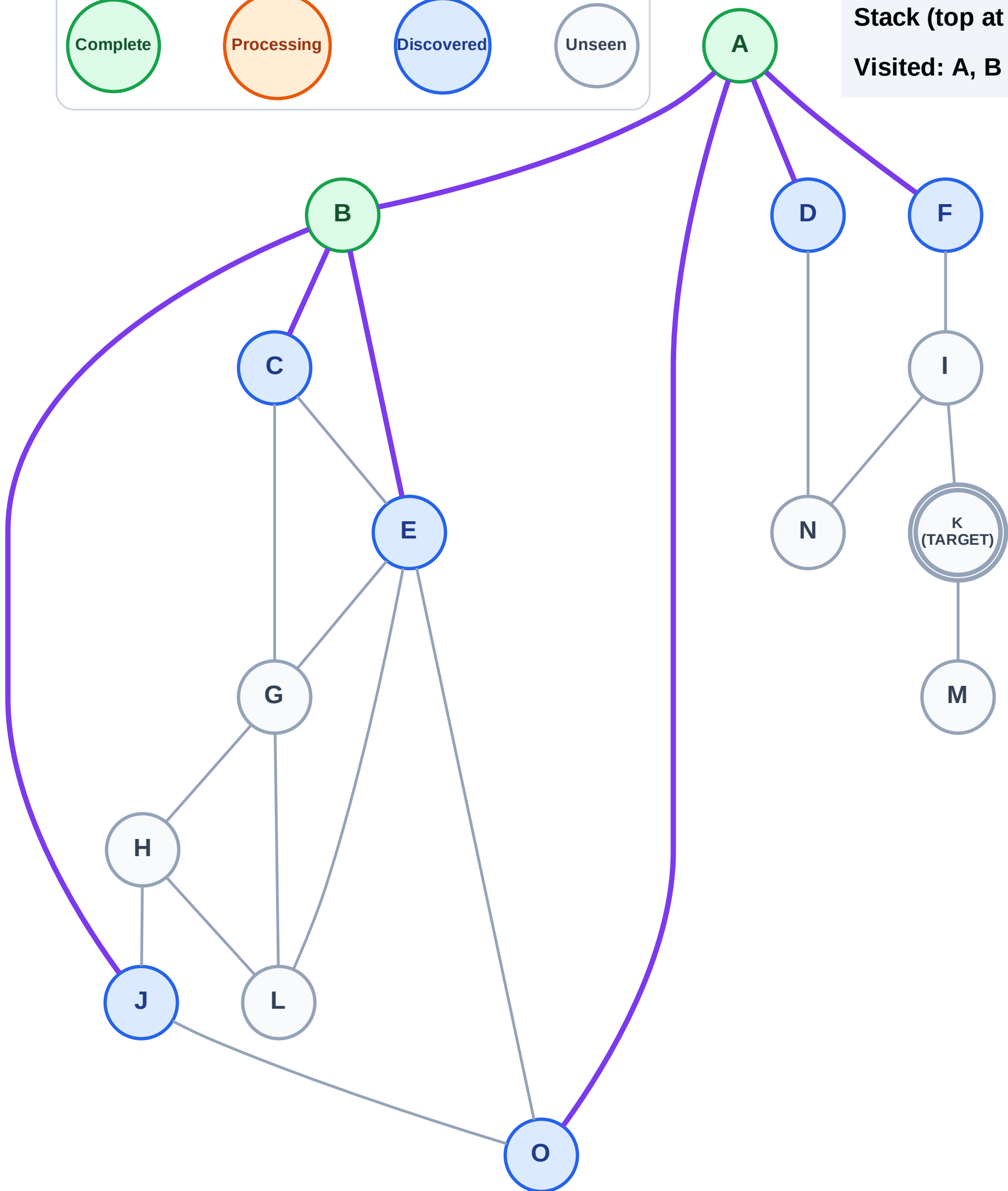
Step 5: Discover J, E, C from B

Legend



Stack (top at right): O | F | D | J | E | C

Visited: A, B



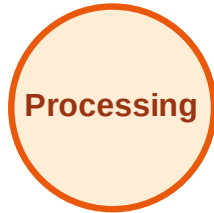
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



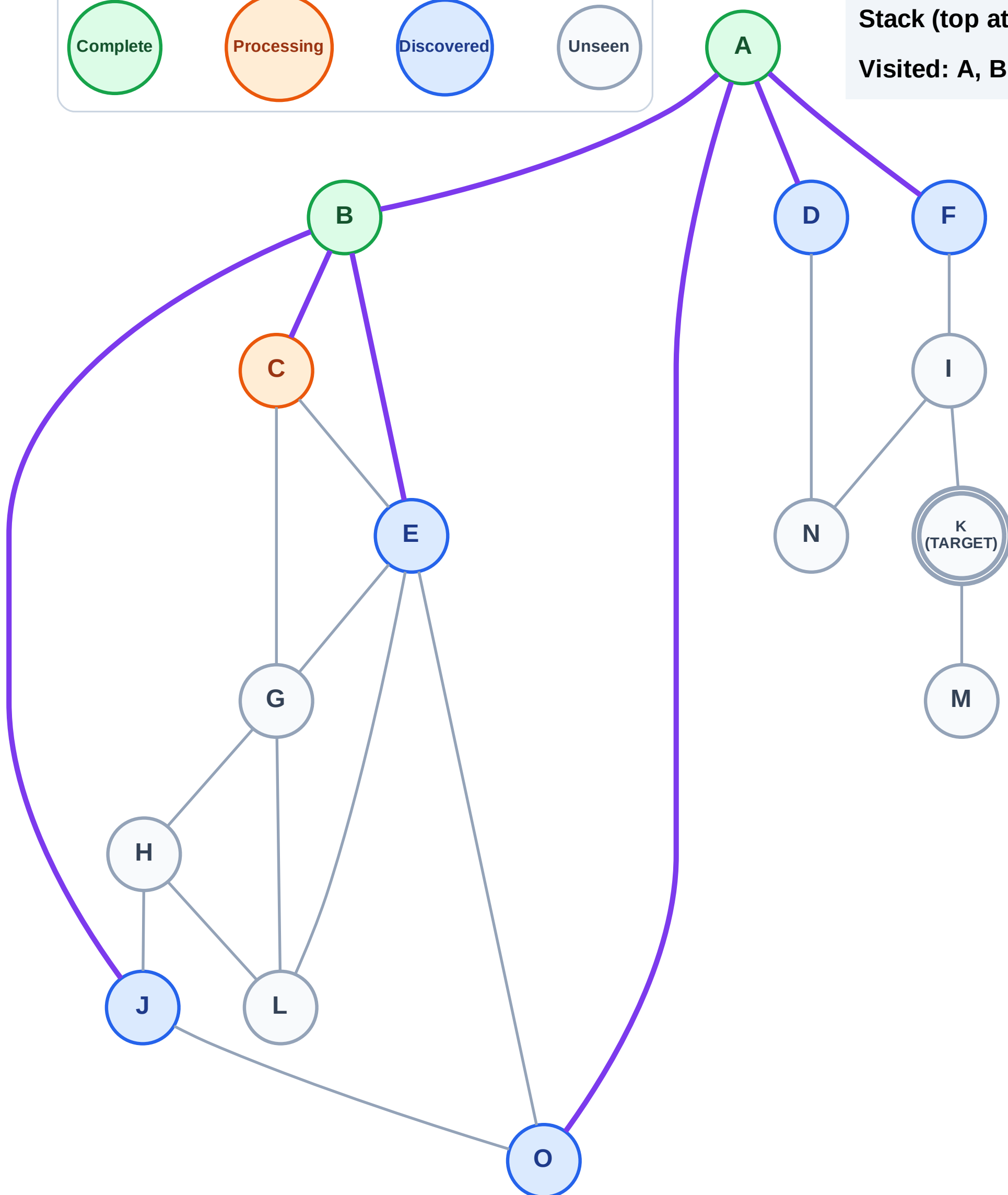
Discovered



Unseen

Stack (top at right): O | F | D | J | E

Visited: A, B



DFS Traversal

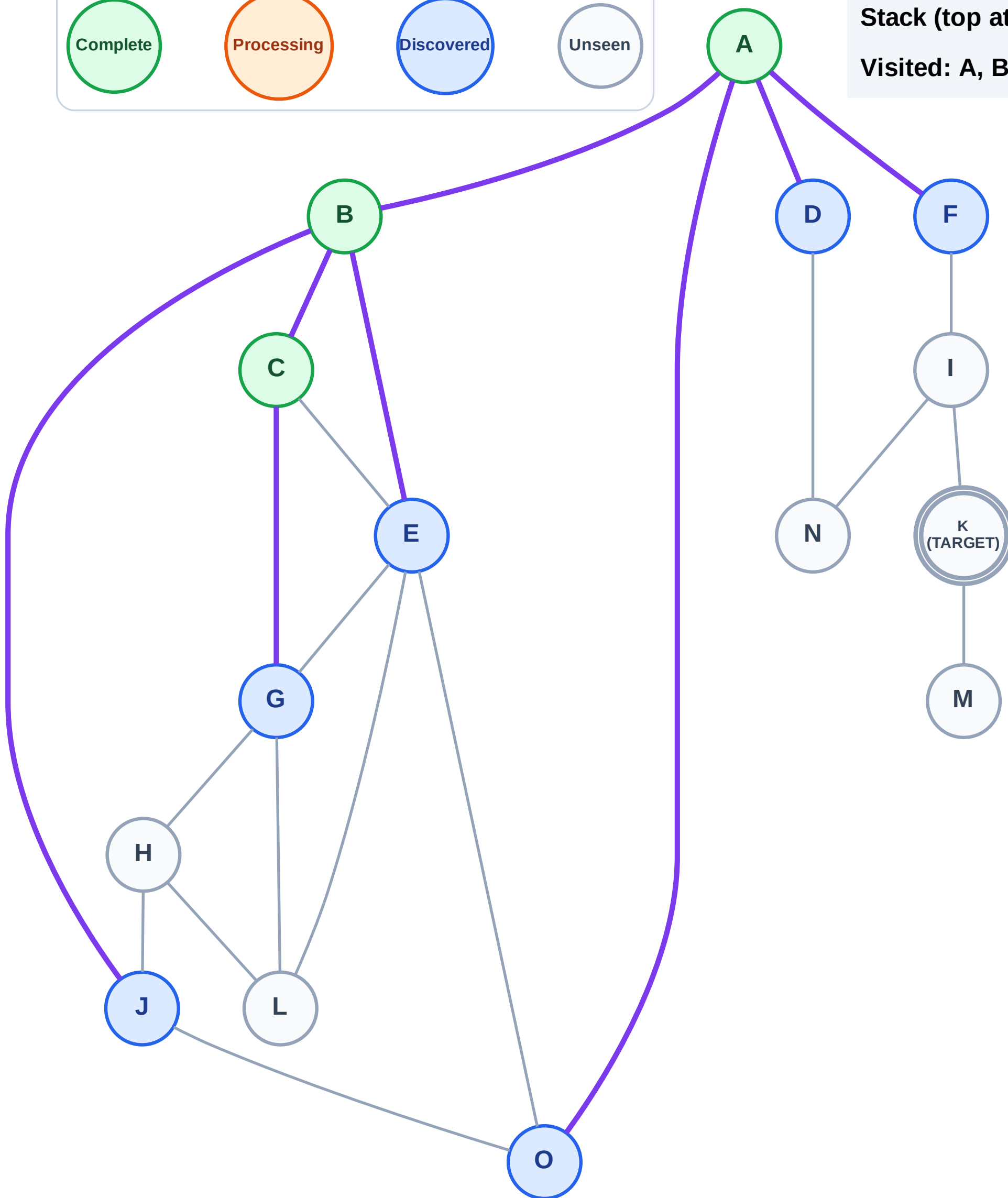
Step 7: Discover G from C

Legend



Stack (top at right): O | F | D | J | E | G

Visited: A, B, C



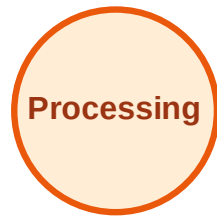
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



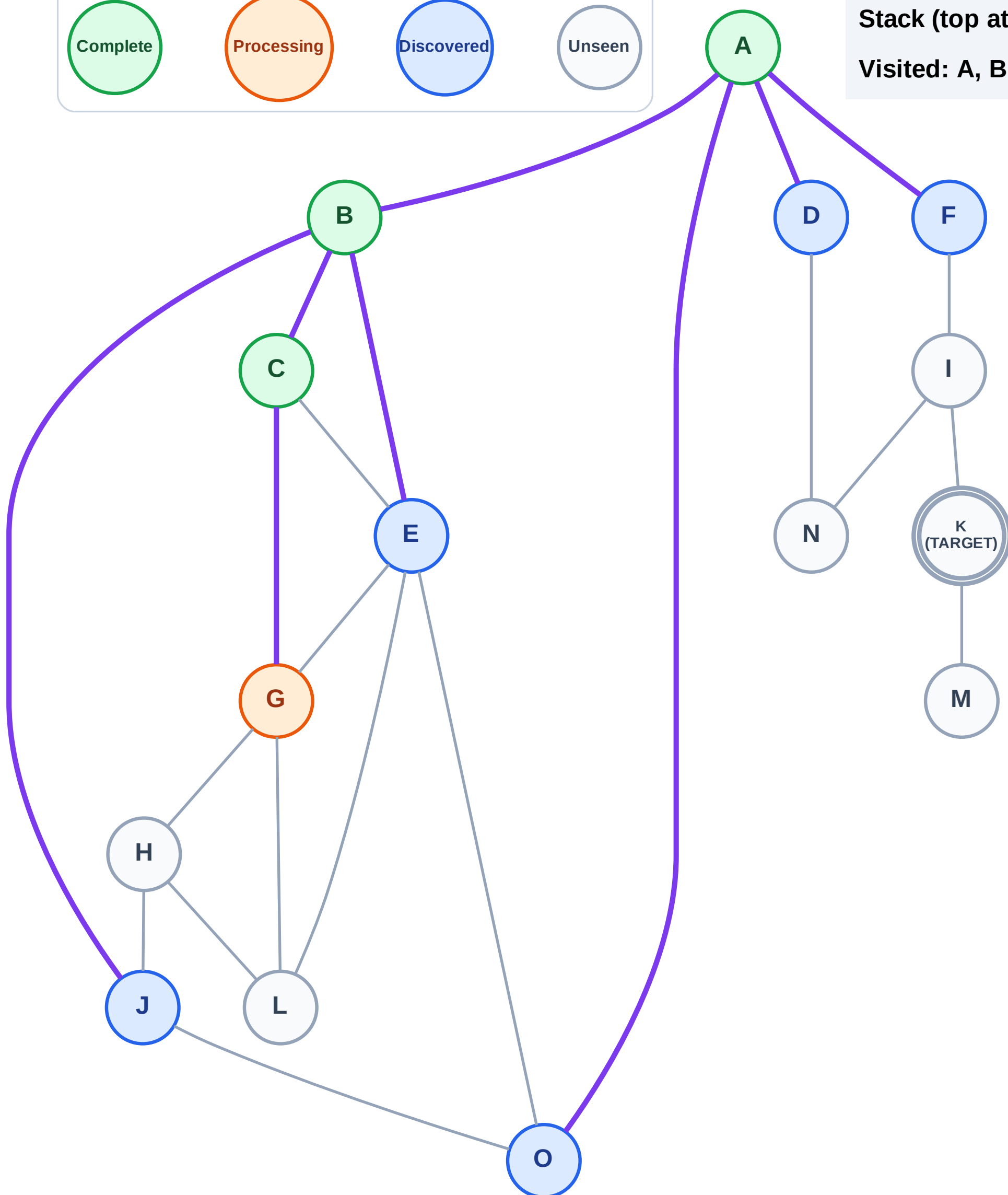
Discovered



Unseen

Stack (top at right): O | F | D | J | E

Visited: A, B, C



DFS Traversal

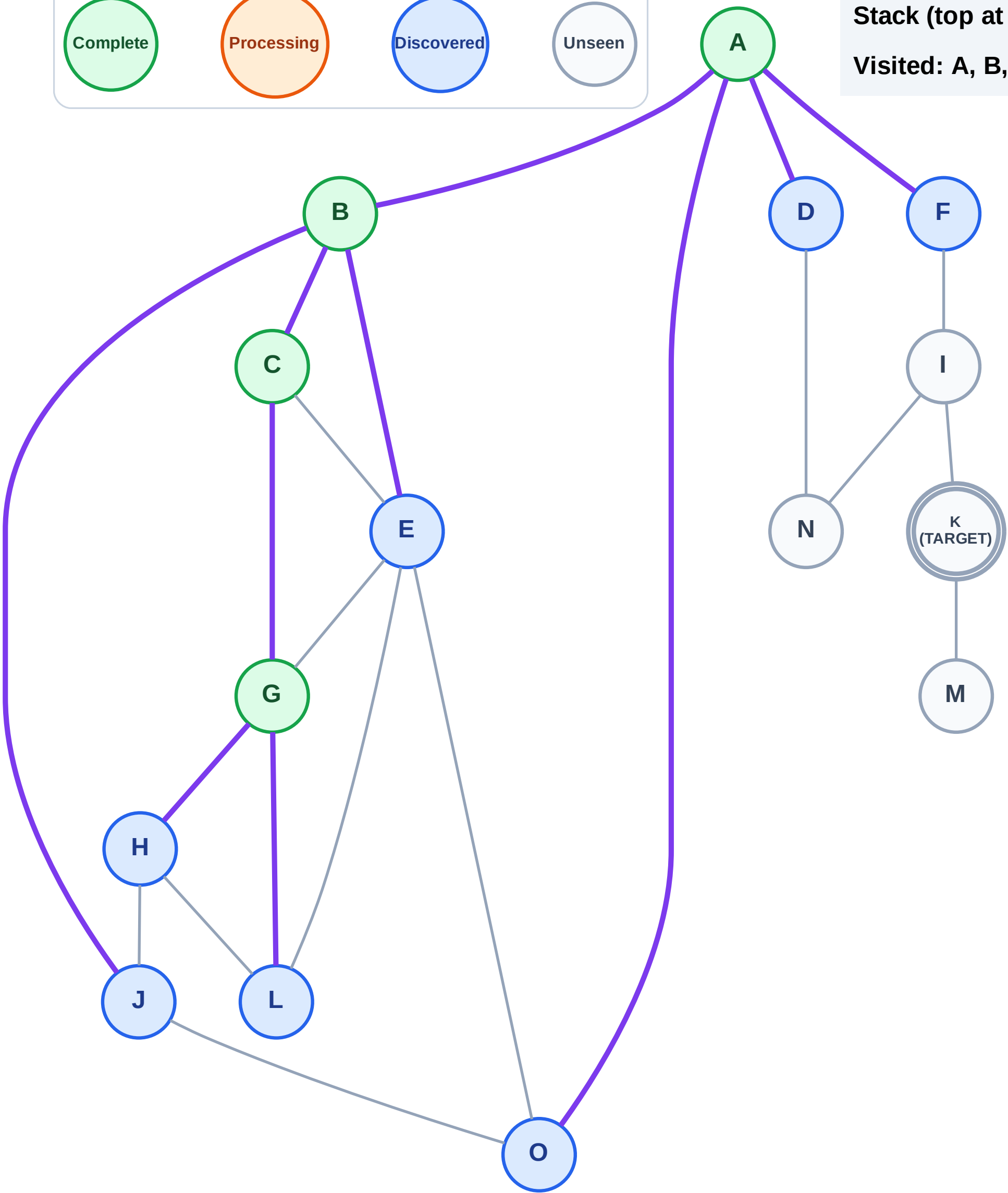
Step 9: Discover L, H from G

Legend



Stack (top at right): O | F | D | J | E | L | H

Visited: A, B, C, G



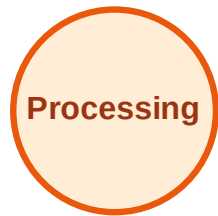
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



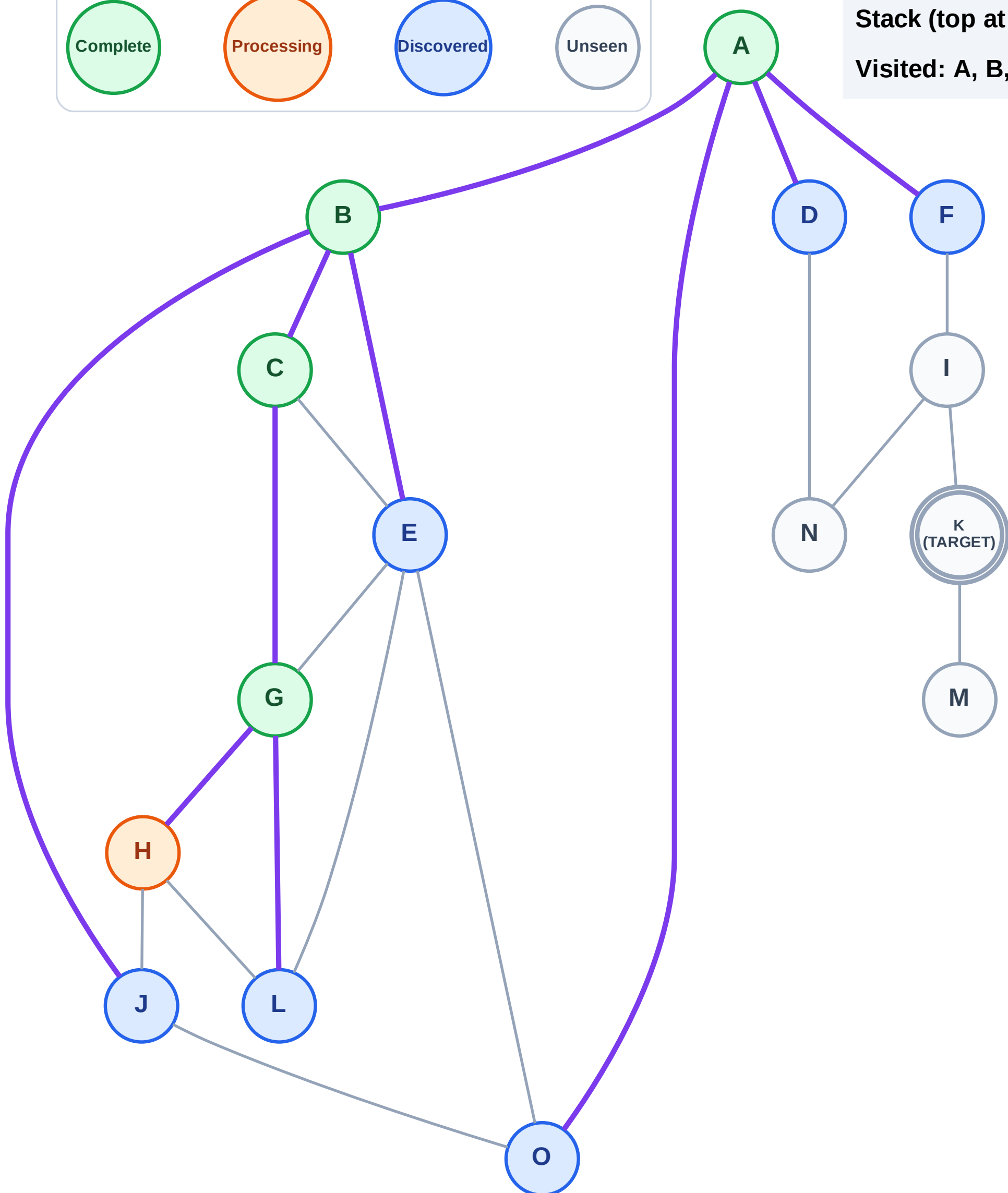
Discovered



Unseen

Stack (top at right): O | F | D | J | E | L

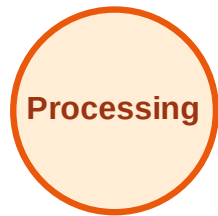
Visited: A, B, C, G



DFS Traversal

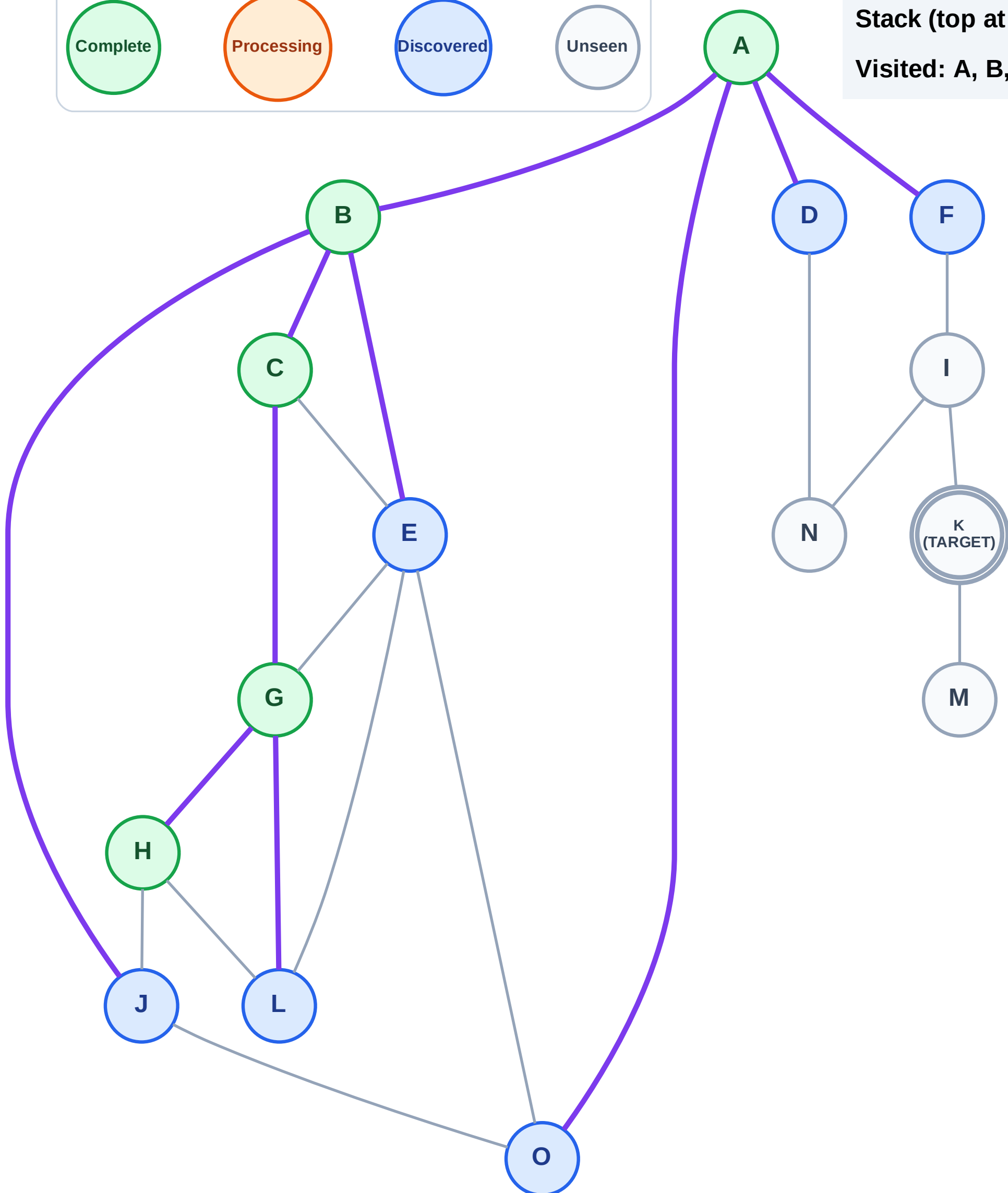
Step 11: No new neighbors from H

Legend



Stack (top at right): O | F | D | J | E | L

Visited: A, B, C, G, H



DFS Traversal

Step 12: Process node L

Legend

Complete

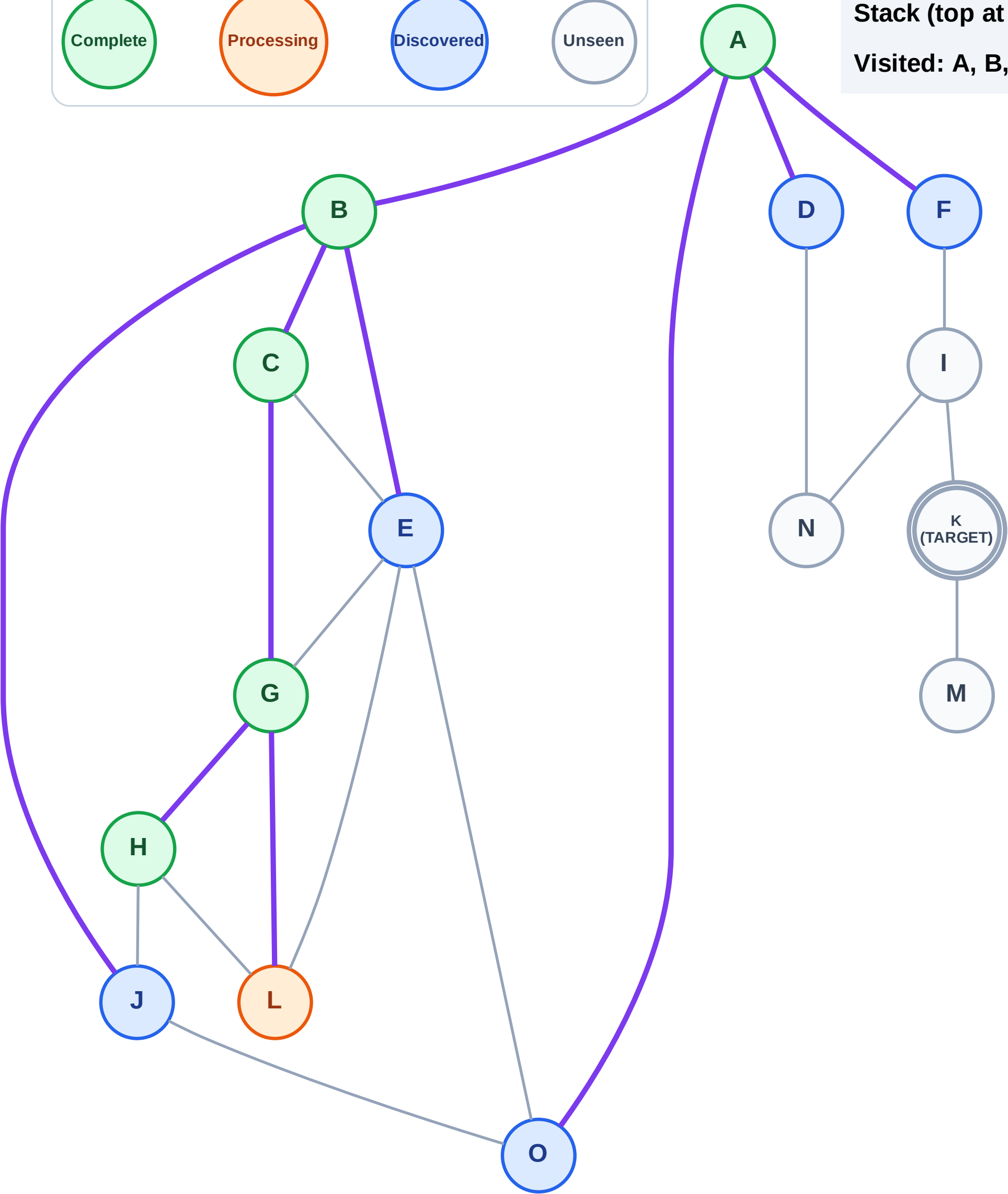
Processing

Discovered

Unseen

Stack (top at right): O | F | D | J | E

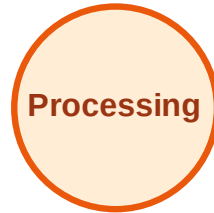
Visited: A, B, C, G, H



DFS Traversal

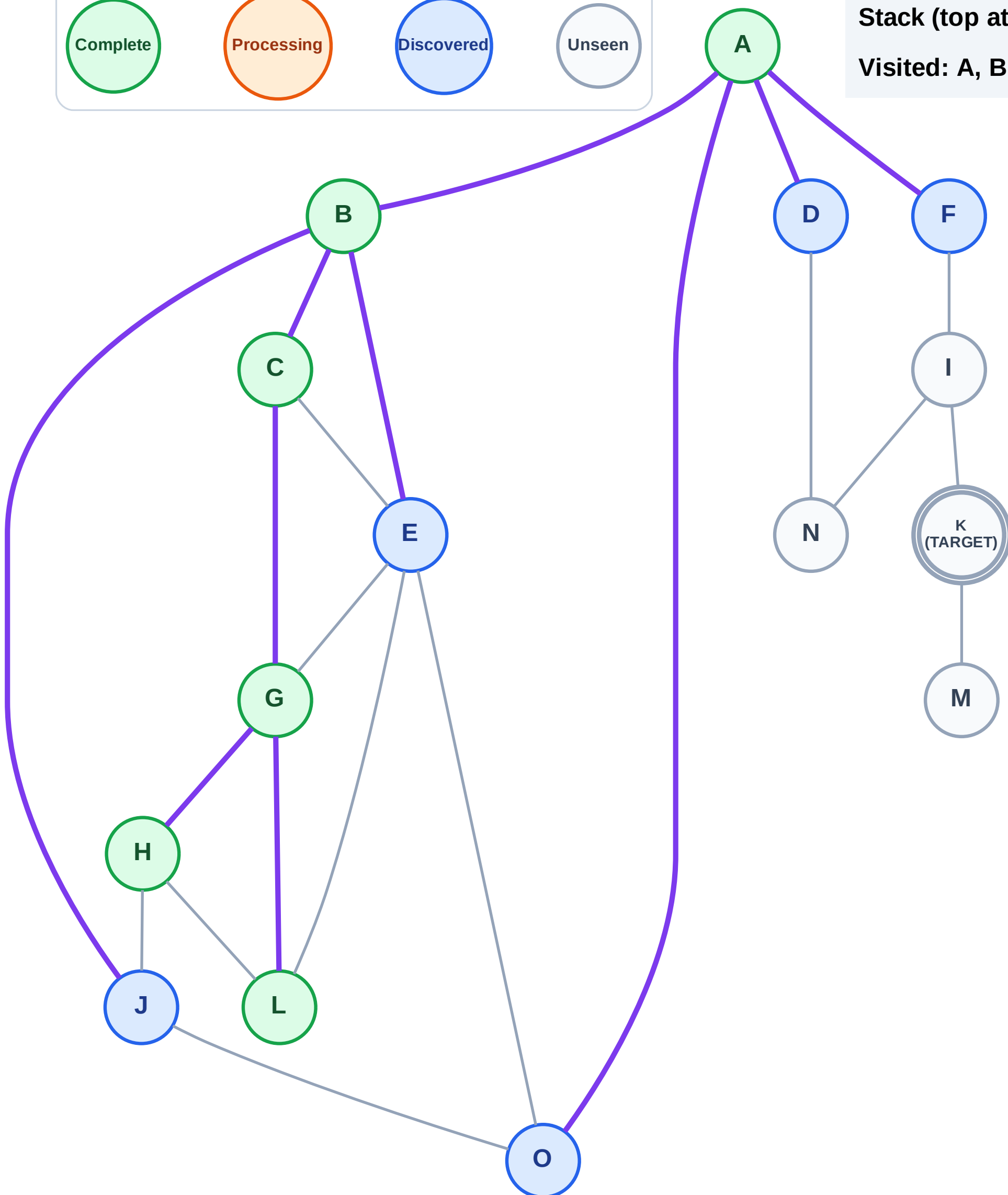
Step 13: No new neighbors from L

Legend



Stack (top at right): O | F | D | J | E

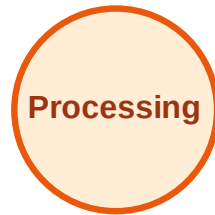
Visited: A, B, C, G, H, L



DFS Traversal

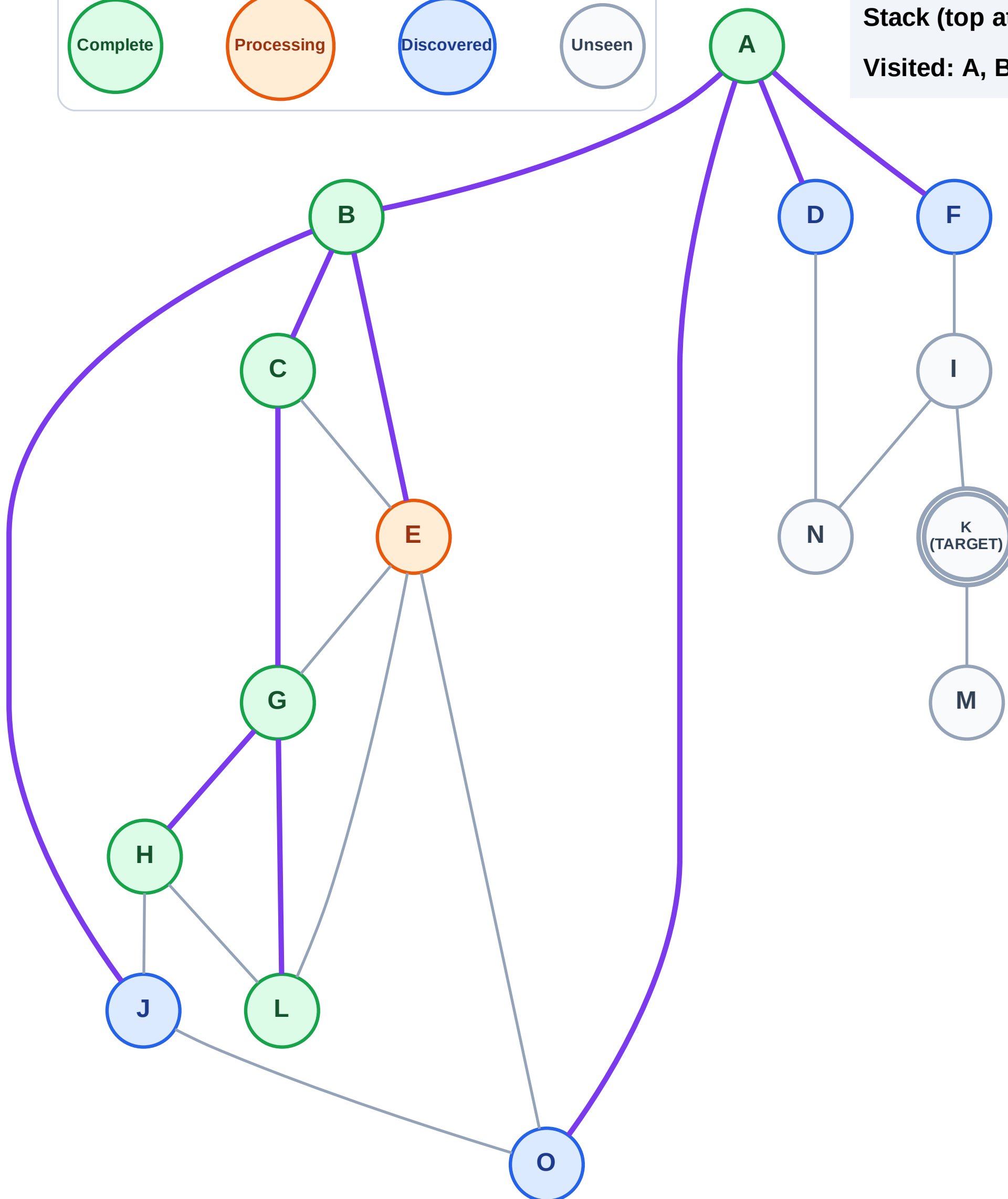
Step 14: Process node E

Legend



Stack (top at right): O | F | D | J

Visited: A, B, C, G, H, L



DFS Traversal

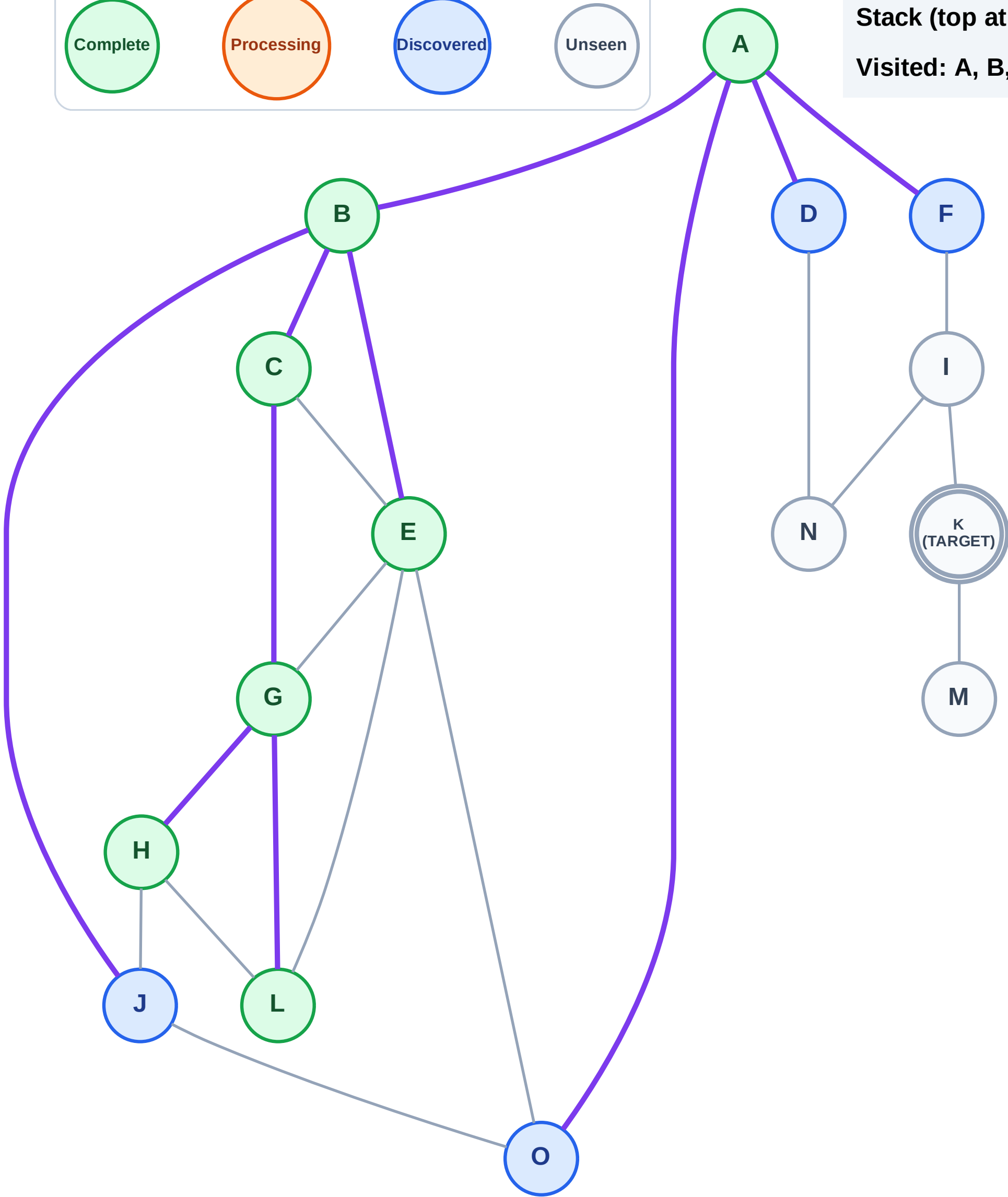
Step 15: No new neighbors from E

Legend



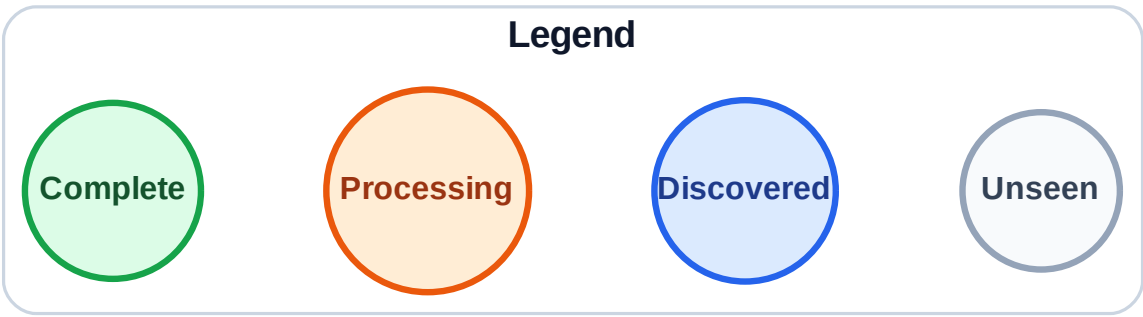
Stack (top at right): O | F | D | J

Visited: A, B, C, E, G, H, L



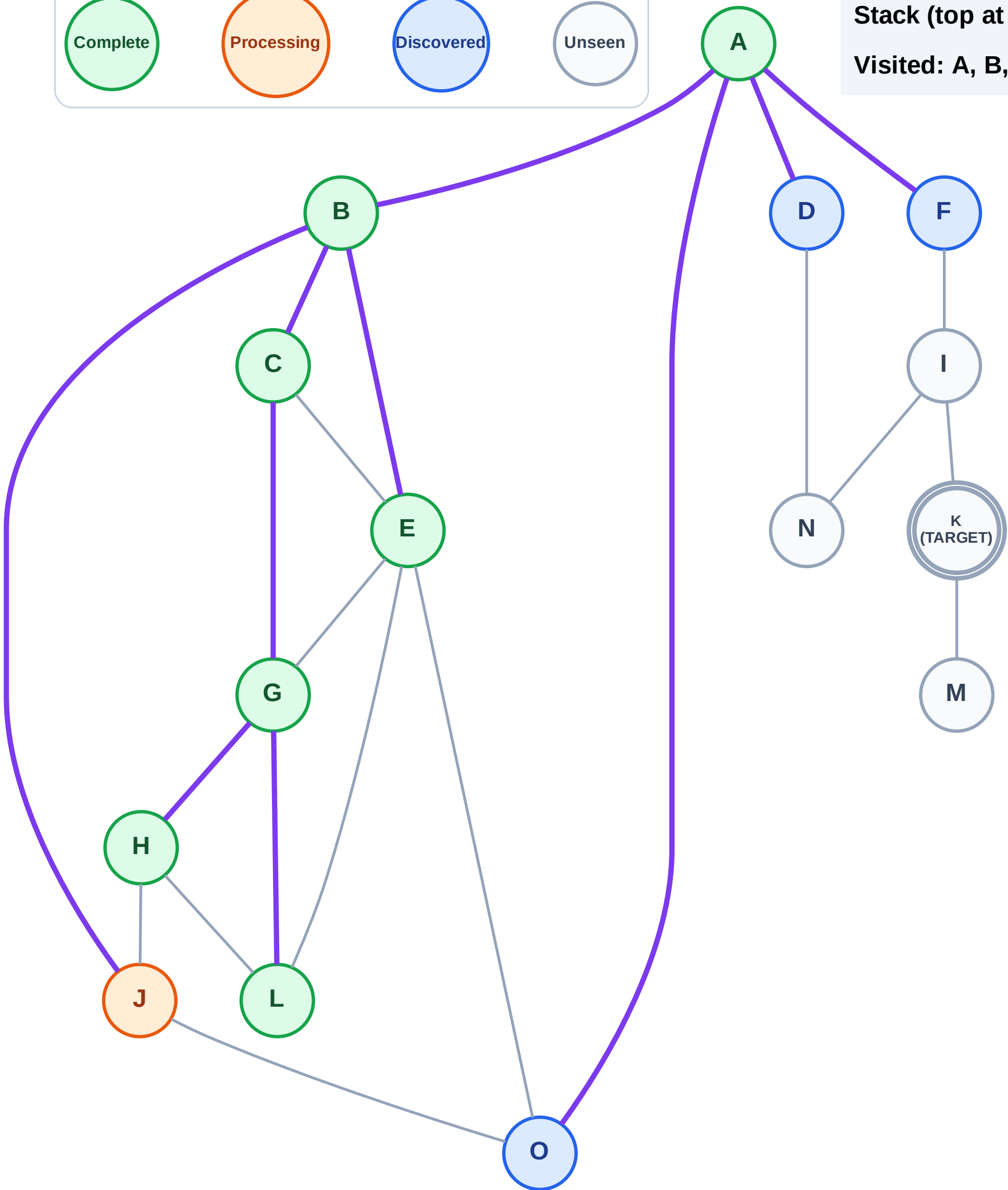
DFS Traversal

Step 16: Process node J



Stack (top at right): O | F | D

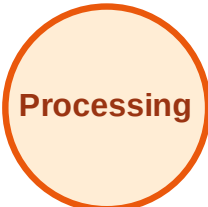
Visited: A, B, C, E, G, H, L



DFS Traversal

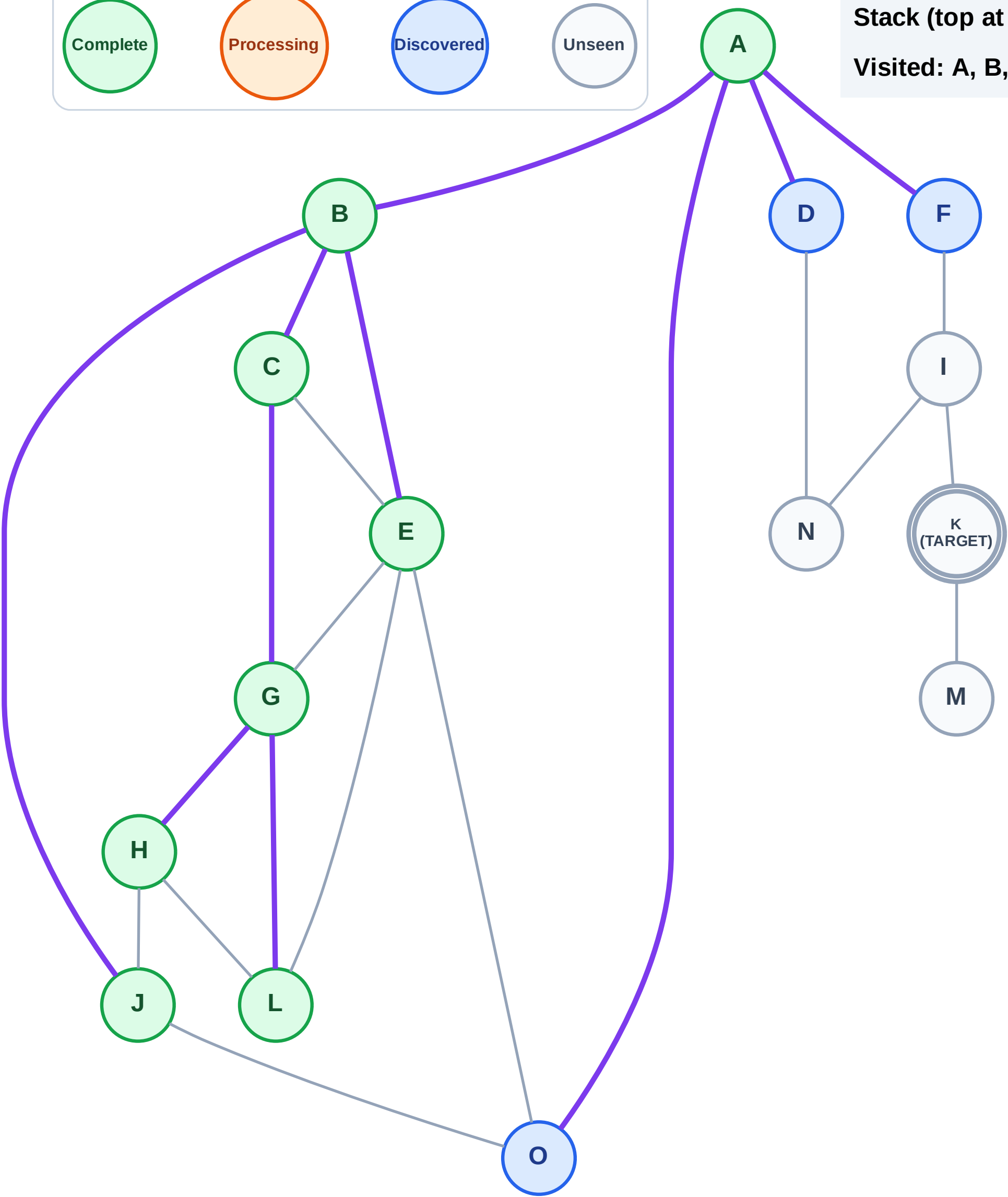
Step 17: No new neighbors from J

Legend



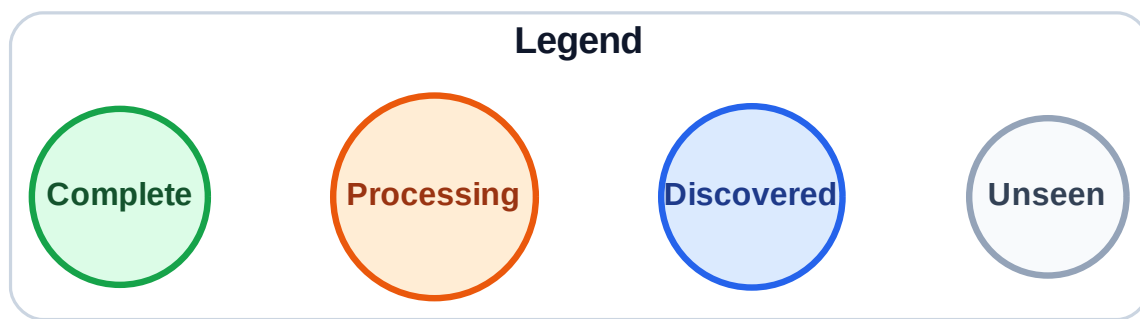
Stack (top at right): O | F | D

Visited: A, B, C, E, G, H, J, L

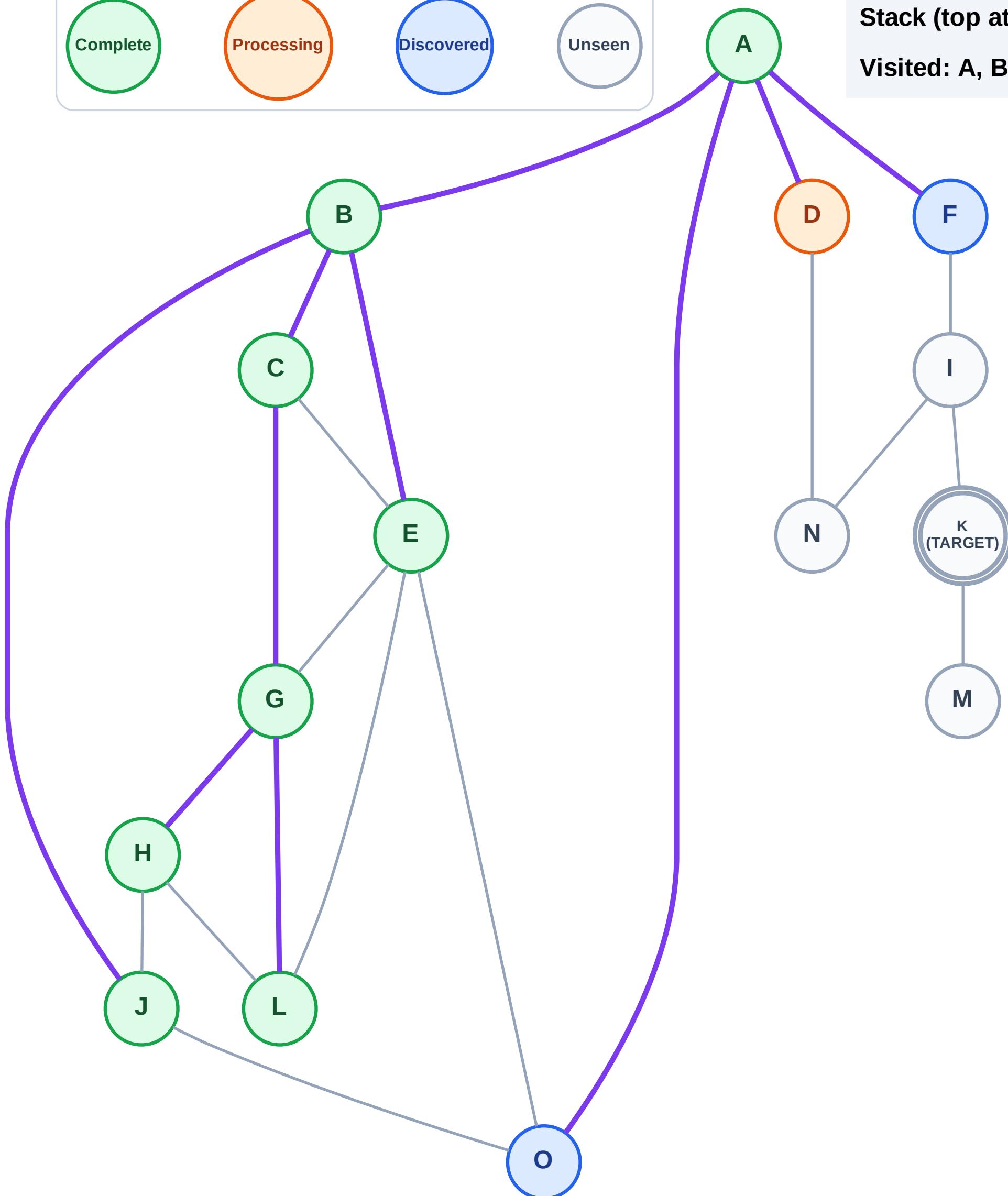


DFS Traversal

Step 18: Process node D



Stack (top at right): O | F
Visited: A, B, C, E, G, H, J, L



DFS Traversal

Step 19: Discover N from D

Legend

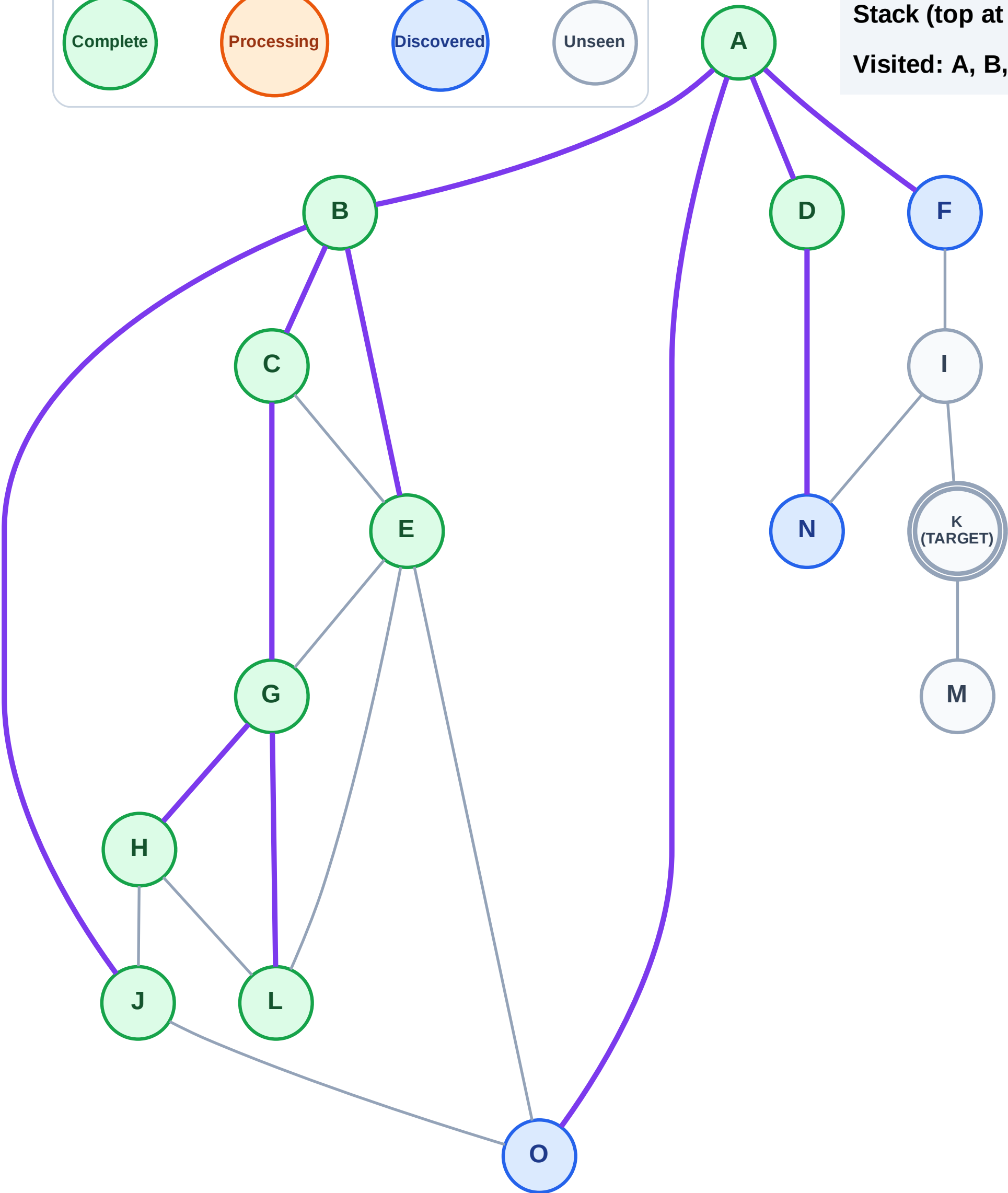
Complete

Processing

Discovered

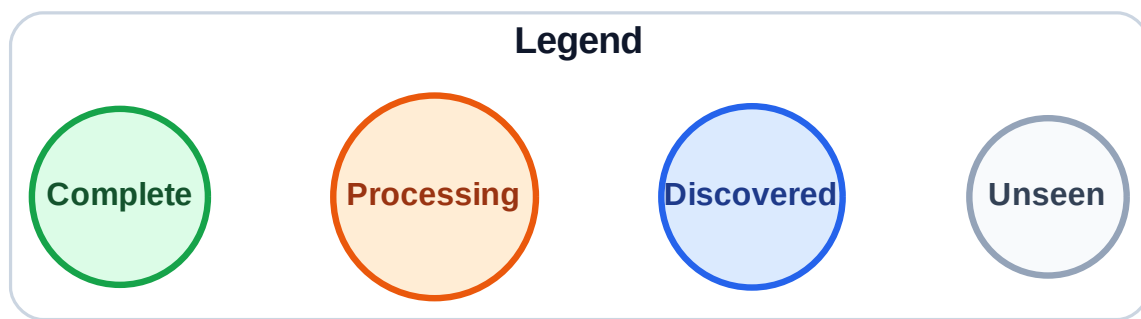
Unseen

Stack (top at right): O | F | N
Visited: A, B, C, D, E, G, H, J, L

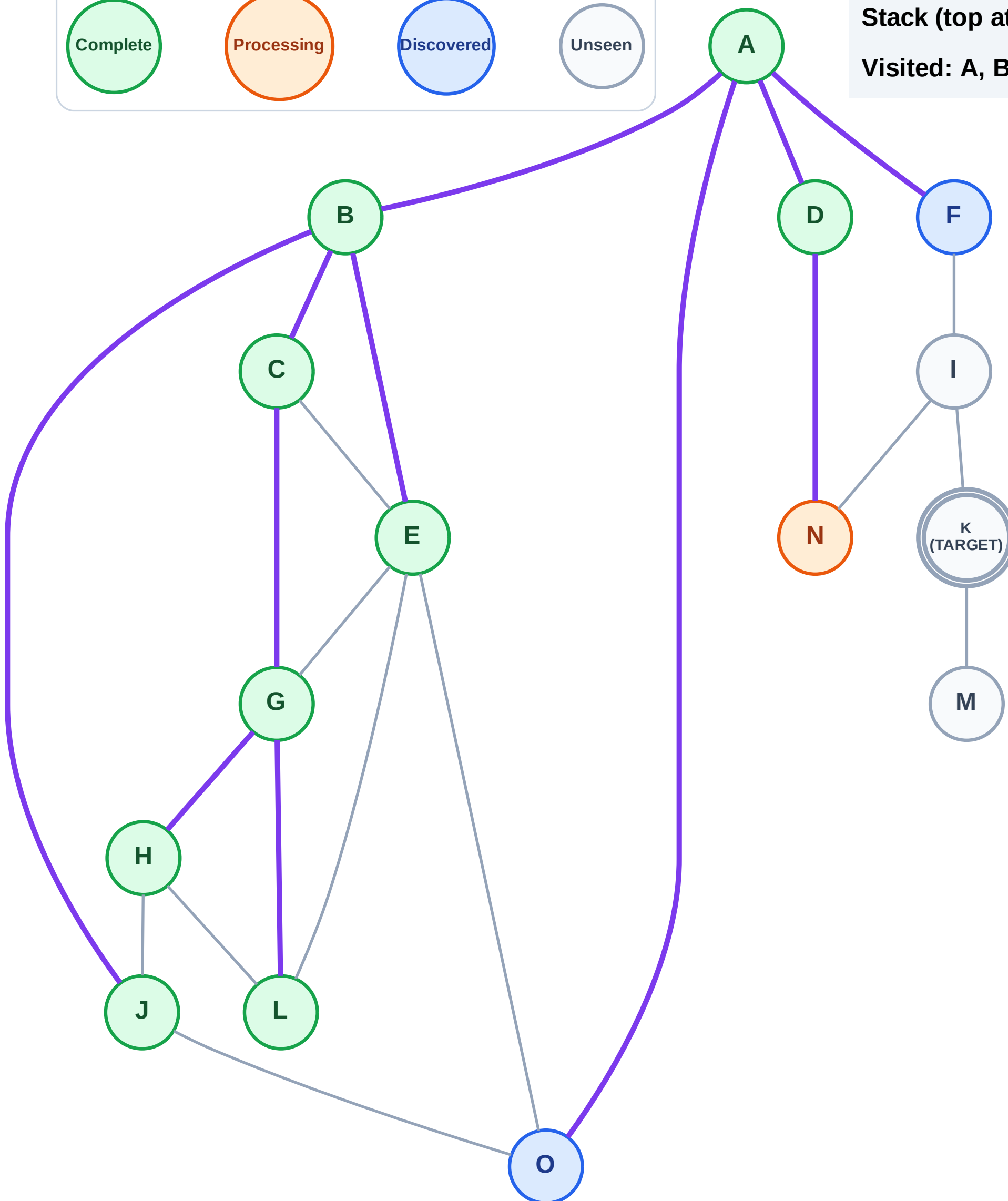


DFS Traversal

Step 20: Process node N



Stack (top at right): O | F
Visited: A, B, C, D, E, G, H, J, L



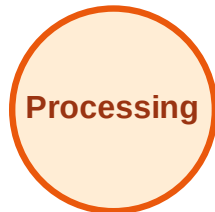
DFS Traversal

Step 21: Discover I from N

Legend



Complete



Processing



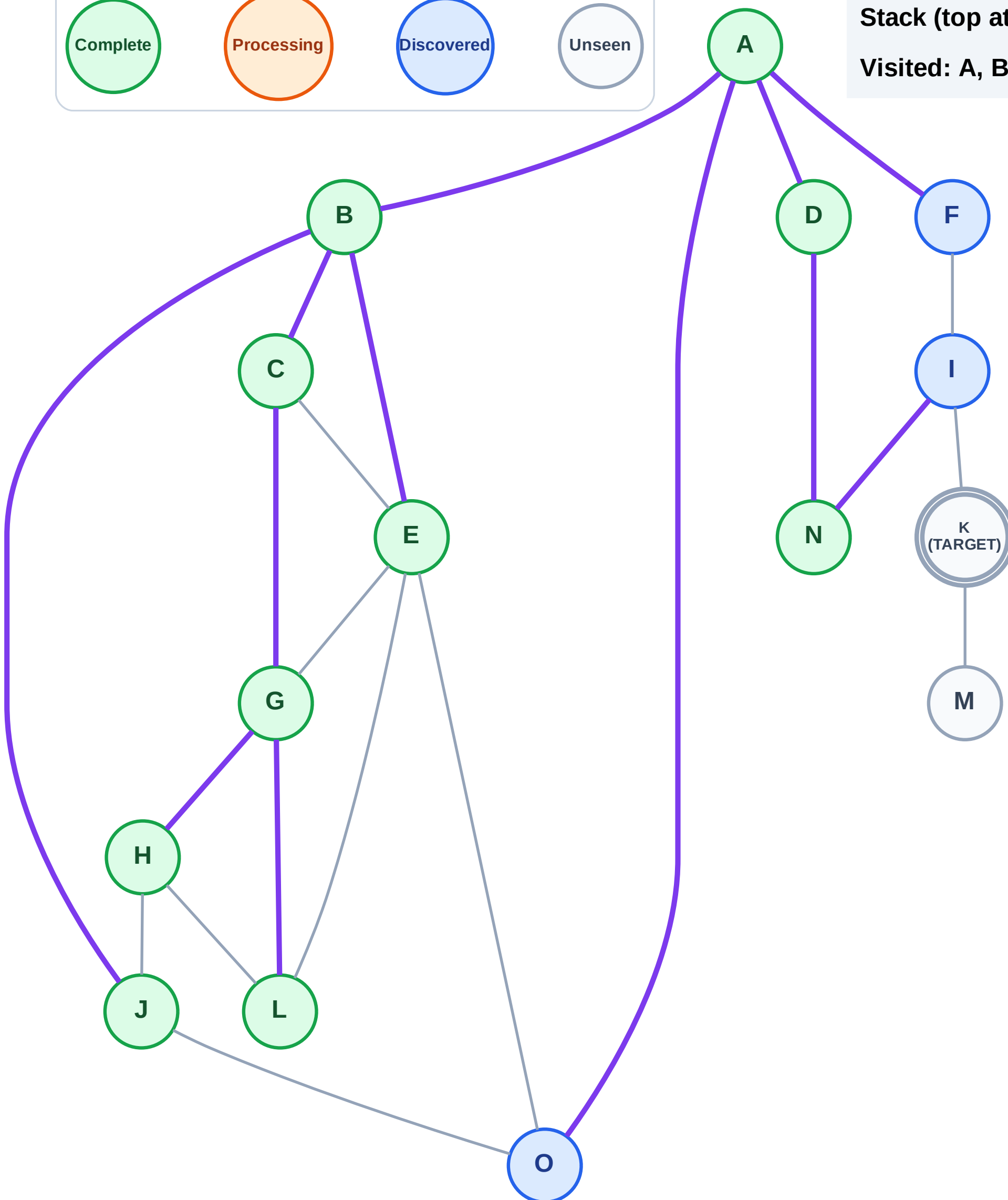
Discovered



Unseen

Stack (top at right): O | F | I

Visited: A, B, C, D, E, G, H, J, L, N



DFS Traversal

Step 22: Process node I

Legend

Complete

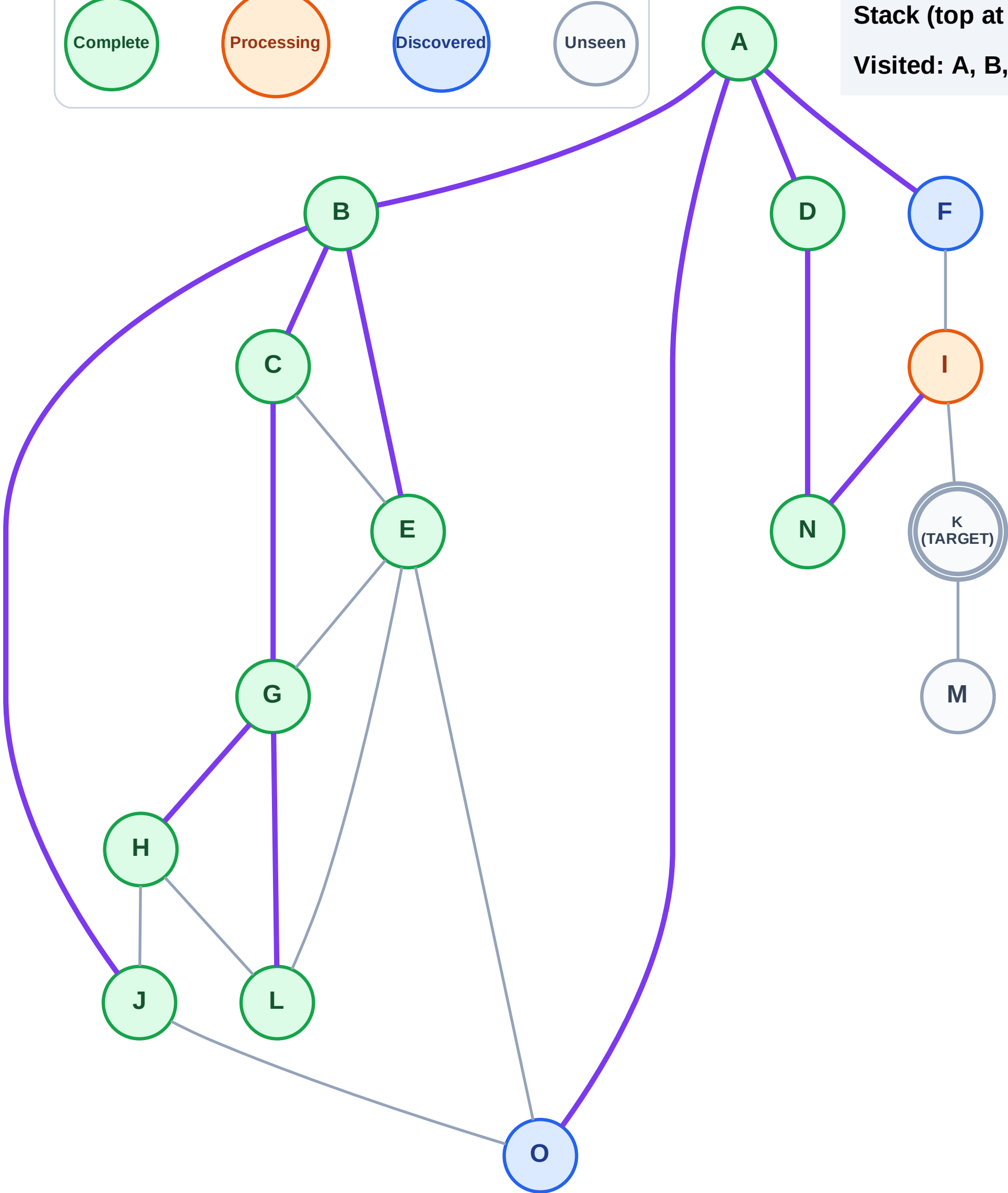
Processing

Discovered

Unseen

Stack (top at right): O | F

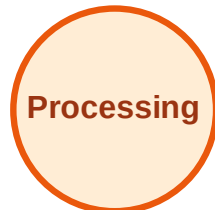
Visited: A, B, C, D, E, G, H, J, L, N



DFS Traversal

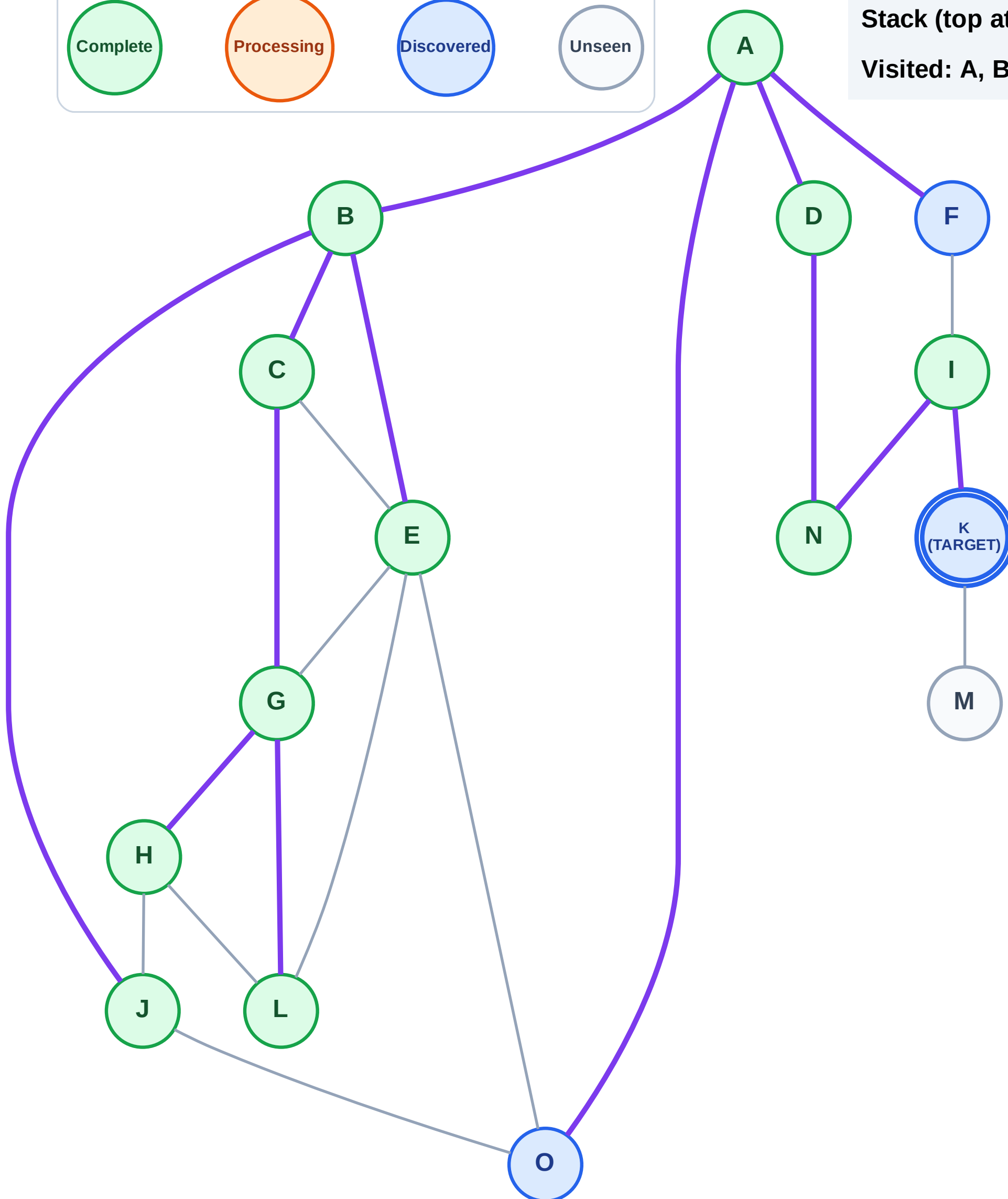
Step 23: Discover K from I

Legend



Stack (top at right): O | F | K

Visited: A, B, C, D, E, G, H, I, J, L, N



DFS Traversal

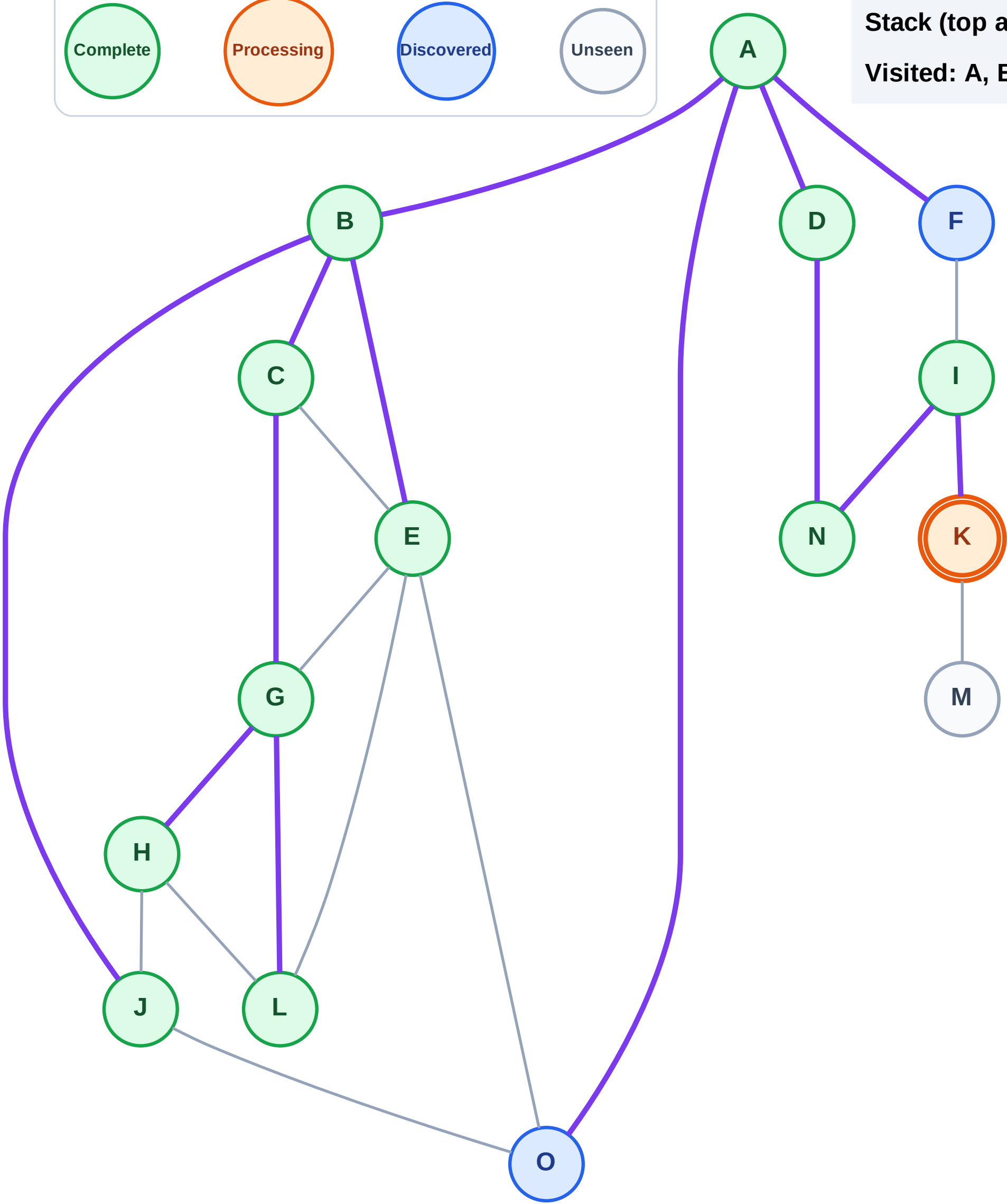
Step 24: Process node K

Legend



Stack (top at right): O | F

Visited: A, B, C, D, E, G, H, I, J, L, N



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (4 hops): A → D → N → I → K

